SRG SIMULATION RECONSTRUCTION

Volume I

Three-Point Attraction and Eight-Body Tetrahedral Dynamics

Historical construction, mathematical transcription and evidence-led experimental reconstruction

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Prepared from three accepted simulation studies and their preserved evidence. No new simulation, scientific checker run, physical assumption or change to the historical equations was made for this volume. GPT accepted the scientific content and closed editorial review. This edition prepares the accepted manuscript for repository publication.

Reader's orientation

Hilmir, these programs begin with a small number of positions and velocities. Their shapes and trails are ways of showing those values. The most useful reconstruction therefore follows two histories at once: what the arrays do, and what the recording and drawing code lets us see.

The first program moves three points using their pairwise directions. Its surviving historical picture is a completed drawing, not a movie. A separately authorized seed-0 reproduction supplies a detailed chronology: two points nearly meet and then separate, while another moves in the opposite direction along much the same visible diagonal. The reproduction clarifies how this rule can make overlapping trails, but does not recover the lost run behind the old picture.

The second program moves eight centers in three dimensions toward a sequentially recalculated group average. Each center is decorated with a fixed tetrahedron. Its movie survives, but its numerical history does not. We can describe specific visible relocations and explain the source mechanisms that permit them without inventing unseen coordinates.

The third program retains that centroid construction and adds a binary response switch. Its two original logs reveal what the movie alone cannot: a switch can first slow a center before reversing its movement; frame labels can restart while the numerical state continues; and a large displayed coordinate jump can accompany a small native step. Its "teleport" diagnostic has a modulo defect that flags many ordinary negative coordinate changes.

This volume connects those findings to exact source statements, equations, rows and images. It preserves unresolved provenance, including the strong but formally unlinked correspondence between the third movie and its shorter log. It completes a first volume of reconstruction, not the wider SRG project.

Abstract

We reconstruct three historical SRG moving-center programs: `recursive_attractor.py`, `3Dtriangles.py`, and `plot_1.py`. The account integrates complete source construction, accepted inspection records, attributed historical graphics and movies, one previously authorized fixed-seed reproduction, and original eight-body CSV evidence. Exact transcriptions preserve sequential update order, wrapped coordinates, carried velocities, a prescribed damped oscillatory gain, componentwise logarithmic damping, and, where present, a mean-speed response factor and stochastic binary interaction sign.

The three-point reproduction establishes a finite near-contact followed by separation, not a recovered historical path-inheritance episode. The no-switch tetrahedral movie supports visible relocation and slower late movement, while lacking the numerical record required for exact displacement claims. The switching program's logs contain 1,092 and 440 eight-body callback blocks despite 200 nominal movie labels. They preserve repeated labels and state continuity through clock resets. At checkable transitions, native increments remain below 0.062 coordinate units; actual coordinate seam events number 22 and 7, whereas the source diagnostic would flag 8,151 and 3,129 body events. The signed contribution can change before velocity reverses.

We derive the source-level drive–damping bookkeeping identity for the recorded kinetic quantity and distinguish it from physical energy conservation. The central conclusion is methodological and mathematical: update ordering, retained state, clock semantics and measurement conventions are necessary to interpret apparent collective behavior. Source-associated media, reproduced data, derived identities and unresolved interpretations remain separate.

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Chapter 1

Scope, historical evidence and method

In ordinary language. The equations, the stored history and the picture answer different questions. A picture can show a line without telling us who arrived first. A program can define a clock without counting every time an animation callback changes the state. We reconstruct these distinctions before interpreting the motion.

1.1 The corpus and the scientific question

This volume asks what three recovered moving-center simulators actually calculated and displayed, and what their surviving evidence establishes. The primary programs are designated S01, S02 and S03 in research order (Table 1.1). These identifiers do not assert a historical creation sequence. The accepted construction studies and the separately authorized three-point replay supply the basis for this synthesis [4, 5, 6, 7, 8]. [C01]

Table 1.1: The selected corpus. Program identity is fixed by the SHA-256 records in Appendix A; media association does not itself certify an exact generating run.

ID and source	Native construction	Surviving evidence used
S01: recursive_attractor.py	Three 2D points, pairwise response, zero initial velocities, 500 sweeps	Attributed historical static PNG; separate seed-0 reproduction with full saved chronology
S02: 3Dtriangles.py	Eight 3D centers, sequential centroid response, random initial velocities	Attributed 2,000-frame historical movie; no certified numerical history for that movie
S03: plot_1.py	Same centroid family plus eight binary response modes, stochastic toggles and diagnostics	Attributed 200-frame movie; original logs A and B; attributed historical body-7 charts

The scope is deliberately narrower than a general SRG theory. We do not merge these programs with fixed-grid systems, later amplitude/neighbor-change models, production physics or memory kernels. Names such as REFU, TGMO, RPCO and GRF are read through their actual implementations. A variable named `force` is not by itself a physical-force law; a function labelled "memory" need not read stored routes. No Newtonian or relativistic field equation, new metric, additional potential or hidden following rule is assumed. [C02]

1.2 Evidence classes and attribution

We use the charter's distinctions throughout:

ORIGINAL
Preserved historical bytes: programs, figures, movies, logs or contemporaneous records.

ATTRIBUTED
A relationship between an artifact and a source or run supported by corroboration, with the remaining uncertainty stated.

OBSERVED
A directly inspected or quantified feature of specified original or reproduced evidence.

REPRODUCED
A new execution of unchanged rules, identified by its initializer and environment; it is not relabelled as lost original data.

DERIVED
A mathematical conclusion from explicit source equations and stated conditions, or an analysis product constructed from existing records.

HYPOTHESIS
An interpretation or research lead not established by the preceding evidence.

MISSING / AMBIGUOUS
An unrecovered identity, state, dependency, chronology or mechanism. Missing evidence is not silently supplied.

These labels can coexist. A movie is an original artifact even when its link to a particular source snapshot is only attributed. Matching a filename, title, geometry and nominal frame rate strengthens source association; it does not recover a random seed, callback sequence or export receipt. Likewise, a derived figure may display original measurements without becoming an original historical plot. [C03]

No original script sets a fixed seed. S01's source makes no file export at all; its historical PNG therefore has an unrecovered saving step. S02 and S03 write fixed movie names relative to the process working directory. Those names can be reused by later executions. S03 gives its CSV a second-resolution timestamp, without a timezone or unique run identifier. The `PLOT_1_` prefix on log A is not produced by that filename expression. Neither current directory placement nor filename dates alone establish which source version generated an artifact. See S01, lines 1–74, S02, lines 110–119 and S03, lines 74–78,158–167. [C04]

1.3 Method and limits of the present synthesis

For the underlying studies, the complete programs were read, their state and execution order were transcribed, relevant original evidence was inspected, and numerical consistency was checked at the declared coverage. This volume reuses those accepted results and their saved verification receipts. It does not rerun the original simulations or their scientific checkers. New operations are editorial: preserving source copies, checking hashes and links, assembling figures, typesetting, inspecting the PDF and packaging. [C05]

For S01's accepted replay, all 500 recorded end-of-sweep position arrays and the final velocity array matched an execution of the unchanged copied source bit for bit in the recorded environment. That comparison does not certify the lost historical PNG or promise bitwise identity on every software stack. For S03, all-row energy, position and incoming-GRF checks coexist with *sampled* full velocity-equation checks. The latter must not be described as an exhaustive audit of every force calculation. [5, 7] [C06]

The canonical study ledger is a navigation record. Its original SIM03 entry was captured while that study was in review. The Volume I work order records GPT's subsequent acceptance within the evidence/provenance qualifications. The ledger is minimally updated accordingly; the accepted study package and its historical in-review snapshot remain unchanged. Scientific and editorial review is complete; publication requires separate authorization. [C07]

1.4 Notation, units and record locators

Positions and velocities are numerical arrays. The programs supply no physical calibration of coordinate length, mass or time. We use "coordinate units" and "model-step units" rather than metres or seconds for the dynamics. Video seconds are actual encoded presentation times; they are not silently identified with model time.

For S01, $n = 0, \ldots, 499$ is the source loop index and S_n the state after n completed sweeps. Event $e = 3n+i+1$ updates body i in sweep $n+1$. For S03, q is a zero-based callback-block index introduced by the accepted analysis in physical CSV order; f_q is the logged `Step` label. Body i at callback q is on physical CSV line $8q + 2 + i$, counting the header as line 1. This locates repeated labels without deduplication. Movie frame indices are zero-based throughout. [C08]

Source locators use the unchanged copies, whose full originals and SHA-256 identities are listed in Appendix A. Major claims carry matrix IDs such as [C08]; the accompanying `CLAIM_EVIDENCE_MATRIX.md` maps each to exact support and caveats. Figures have independent provenance in `FIGURE_AND_MEDIA_REGISTER.md`. These registers are part of the scientific review package, not a substitute for the explanations below.

Chapter 2

Simulation 01: three interacting points

In ordinary language. Three points start at different random locations, at rest. Each point responds to the other two, adds that response to its retained velocity, damps the result, and moves. The points take turns. Only after all 500 rounds does the program draw their complete trails. The trail is a record, not an extra moving object or a route the points consult.

2.1 Entities, initialization and retained state

The complete 74-line `recursive_attractor.py` imports NumPy and Matplotlib, defines three helpers, performs the whole numerical loop at module level, then makes a static plot. There is no main guard or animation callback. Importing the file would execute the simulation; no such import was made for this volume. S01, lines 1–74; [4]. [C09]

There are three position vectors $x_i \in [-1,1)^2$ and three velocity vectors $v_i \in \mathbb{R}^2$, giving twelve real-valued state entries. The source initializes six coordinates by `np.random.rand(3,2)*2-1`, with no seed statement, and sets all six velocity entries to zero. It sets $N = 3$, $\Delta t = 0.05$ and 500 sweeps. There are no masses, radii, collision surfaces, binary modes or permanent leader/follower assignments. The legend later labels array indices $0, 1, 2$ as Attractors $1, 2, 3$. S01, lines 5–14,66–71. [C10]

The current arrays carry state between sweeps. The initially empty `position_log` stores observations for drawing. Neither that log nor its earlier entries are read by the movement rule. A point's velocity therefore retains consequences of previous interactions, but the program does not contain an explicit remembered itinerary. S01, lines 14,38–62. [C11]

2.2 The exact operators

The source's function H is

$$H(t) = 2e^{-0.5t}\cos t + e^{-0.3t}\sin t + 1.5e^{-0.7t} + e^{-0.4t}. \tag{2.1}$$

Its four amplitude/decay pairs are $(2, 0.5)$, $(1, 0.3)$, $(1.5, 0.7)$ and $(1, 0.4)$. It depends only on its time argument. The name "REFU memory curve" in the source comment does not add a dependence on recorded positions. S01, lines 16–27. [C12]

Define componentwise wrapping and damping by

$$\mathcal{W}_2(z) = ((z+1) \bmod 2) - 1, \tag{2.2}$$

$$\mathcal{R}(u)_a = \frac{u_a}{1 + \log(1 + |u_a|)}. \tag{2.3}$$

Equation (2.2) is TGMO, used both on relative displacements and on new positions. Its range is half-open; an exact half-period tie selects -1 in that coordinate. Equation (2.3) acts separately on components, not on the Euclidean norm of a whole vector. S01, lines 29–35. [C13]

For one sweep n, let x_i^n, v_i^n denote the incoming state and x_i^{n+1}, v_i^{n+1} the values after body i has been updated. Before updating body i, define the actually available positions

$$\widetilde{x}_j^{n,i} = \begin{cases} x_j^{n+1}, & j < i, \\ x_j^n, & j \geq i. \end{cases} \tag{2.4}$$

For $i = 0, 1, 2$ in that order, the exact mathematical transcription is

$$t_n = 0.05n, \qquad h_n = H(t_n), \tag{2.5}$$

$$d_{ij}^{n,i} = \mathcal{W}_2(\widetilde{x}_j^{n,i} - x_i^n), \qquad j \neq i, \tag{2.6}$$

$$F_i^n = \sum_{j \neq i} \frac{d_{ij}^{n,i}}{\|d_{ij}^{n,i}\|_2 + 0.001}, \tag{2.7}$$

$$u_i^n = v_i^n + 0.05 h_n F_i^n, \tag{2.8}$$

$$v_i^{n+1} = \mathcal{R}(u_i^n), \qquad x_i^{n+1} = \mathcal{W}_2(x_i^n + 0.05 v_i^{n+1}). \tag{2.9}$$

The shared h_n is evaluated once per sweep. The regularizer is added *after* taking the norm; it is not a squared-distance softening. At coincidence the corresponding displacement contribution is zero. No inverse-square law, potential or physical unit is supplied by these expressions. S01, lines 38–56. [C14]

2.3 One sweep in execution order

At the beginning of a sweep, the source creates an empty `new_positions` list. For each body it resets a two-component interaction sum, visits the other indices, skips itself, wraps each difference and adds its regularized direction. It then performs velocity addition, componentwise damping and position wrapping, and appends a *copy* of the resulting position. S01, lines 42–57.

Table 2.1: Position dependencies during one S01 sweep. A common clock does not imply simultaneous numerical updates.

Body updated	First other position	Second other position
0	Old body 1	Old body 2
1	New body 0	Old body 2
2	New body 0	New body 1

The list is an observation buffer, not an array of proposals committed simultaneously. Earlier mutations are already visible to later bodies. After the third move, the three copied positions form an end-of-sweep snapshot because later bodies do not overwrite earlier rows. The final log has shape $(500, 3, 2)$ and excludes the initializer. It stores neither velocities nor within-sweep comparisons. S01, lines 43–62. [C15]

The first stored state was produced with $t = 0$; the last with $t = 24.95$. Counting 500 multipliers of 0.05 gives 25 model-step units, but no physical-time calibration. The program stops because its loop ends, not because it detects equilibrium. A zero initial velocity is not restored after each sweep. S01, lines 5–11,38–59. [C16]

2.4 What the historical drawing establishes

After simulation, one plot call per body connects its 500 recorded positions into 499 line segments. The script supplies title, legend, grid, equal axis scaling and a final `plt.show()`. It supplies no `savefig`, CSV export, movie writer, temporal arrows or initial-state marker. The saving method for the surviving `attractors.png` is missing. S01, lines 64–74. [C17]

Figure 2.1 is the original attributed image. Its exact title and three traces match the source construction. Closely aligned upper curves, separate turns and long near-vertical central connections are visible. Those features do not disclose the order in which the points visited a location. Since wrapped samples are connected directly, the source permits a seam crossing to produce a long flat-plot connector. Without the generating numerical log, that is an explanation the source permits, not a certified diagnosis of each line in the old PNG. [4] [C18]

The plotted line is also a discrete drawing convention. It is not a continuous-time solution supplied by the code. Neither overlapping segments nor the apparent merging of colors proves simultaneous occupancy, exact coincidence, a bound pair or route inheritance. The original seed and ordered history were not recovered in the bounded construction study. That negative recovery result does not prove no such evidence exists elsewhere. [C19]

2.5 The separately accepted seed-0 reproduction

The subsequent replay study was explicitly authorized as one new 500-sweep configuration using legacy NumPy seed 0. Its initializer is shown in Table 2.2. The historical rules, zero velocities, arithmetic order, clock and wrapping were preserved. The accepted environment was Python 3.11.15, NumPy 2.4.4, Matplotlib 3.10.9 and float64 arrays, with Agg rendering. [5] [C20]

Table 2.2: The reproduced initializer, not the missing historical initializer. All initial velocities are $(0, 0)$. Full floating-point values and hexadecimal representations remain in replay/initial_state.json.

Body	X	Y
0	0.0976270078546495	0.43037873274483895
1	0.20552675214328775	0.08976636599379373
2	−0.15269040132219058	0.29178822613331223

The accepted harness recorded 501 sweep-boundary states including S_0, 1,500 ordered point events, incoming and outgoing velocities, comparisons, native increments and integer wrap shifts. A single same-seed reference execution of the unchanged copied source supplied a fidelity comparison: all 3,000 logged position components and six final velocity components matched bit for bit. Intermediate reference velocities were not compared at every sweep. The present volume reads that receipt; it performs neither execution again. [C21]

Figure 2.2 is therefore labelled reproduced, not original. Its long shared diagonal belongs to this known seed. The very different historical picture in Figure 2.1 cannot be assigned this initializer or chronology. [C22]

2.5.1 The opening updates

The first inspection window was fixed as events 1–9, the first three sweeps. In event 1, body 0 moves by approximately $(-0.005728, -0.012602)$. Body 1 then sees its changed coordinates and moves by approximately $(-0.010860, +0.012595)$. Body 2 sees both updated positions and moves by approximately $(+0.014857, -0.000239)$. These are saved records, not new calculations

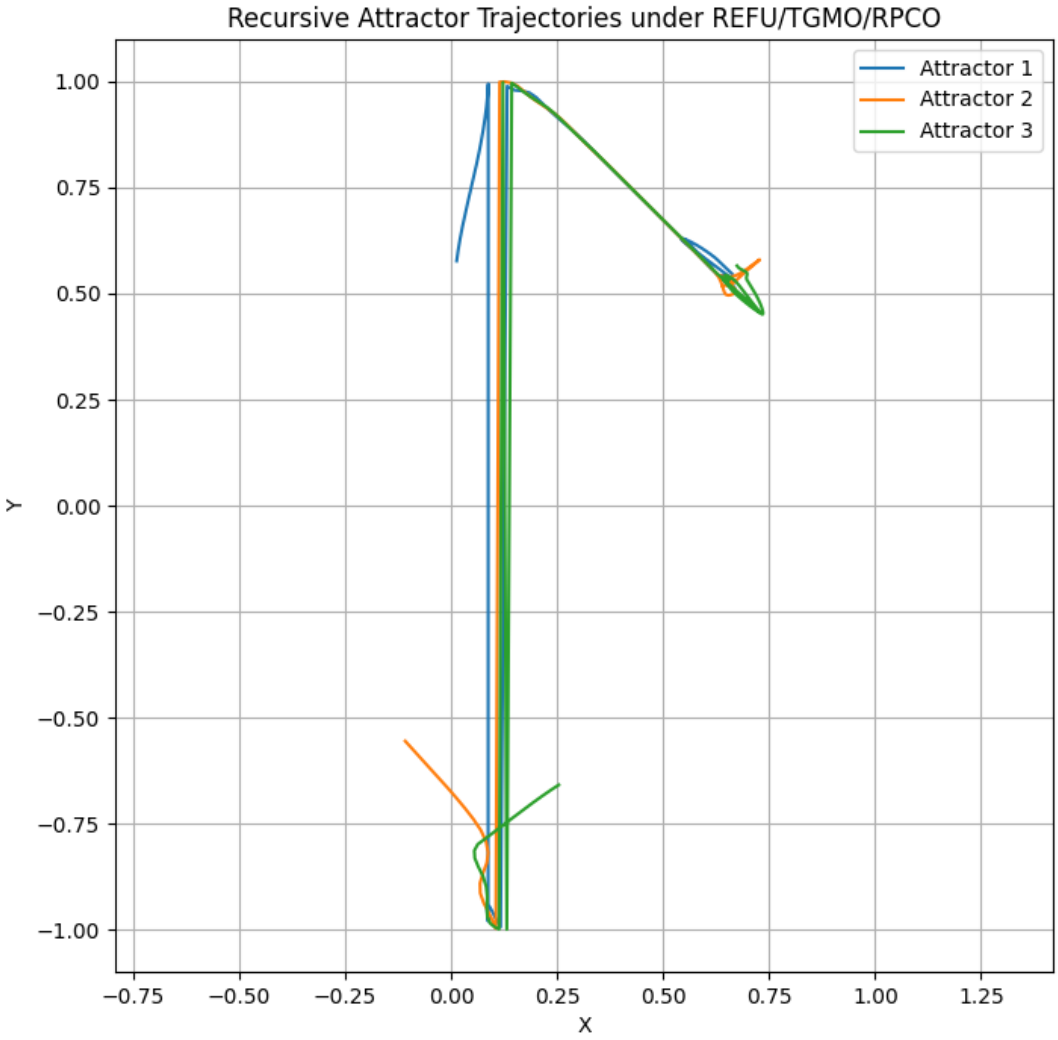


Figure 2.1: ORIGINAL / ATTRIBUTED. The historical attractors.png, preserved in Construction 01. Axes are X/Y coordinate values; the three connected traces have no time arrows or unique event labels. Its title and geometry support source association, but the initializer, ordered log and saving method are missing. This is not the seed-0 reproduction. [4]

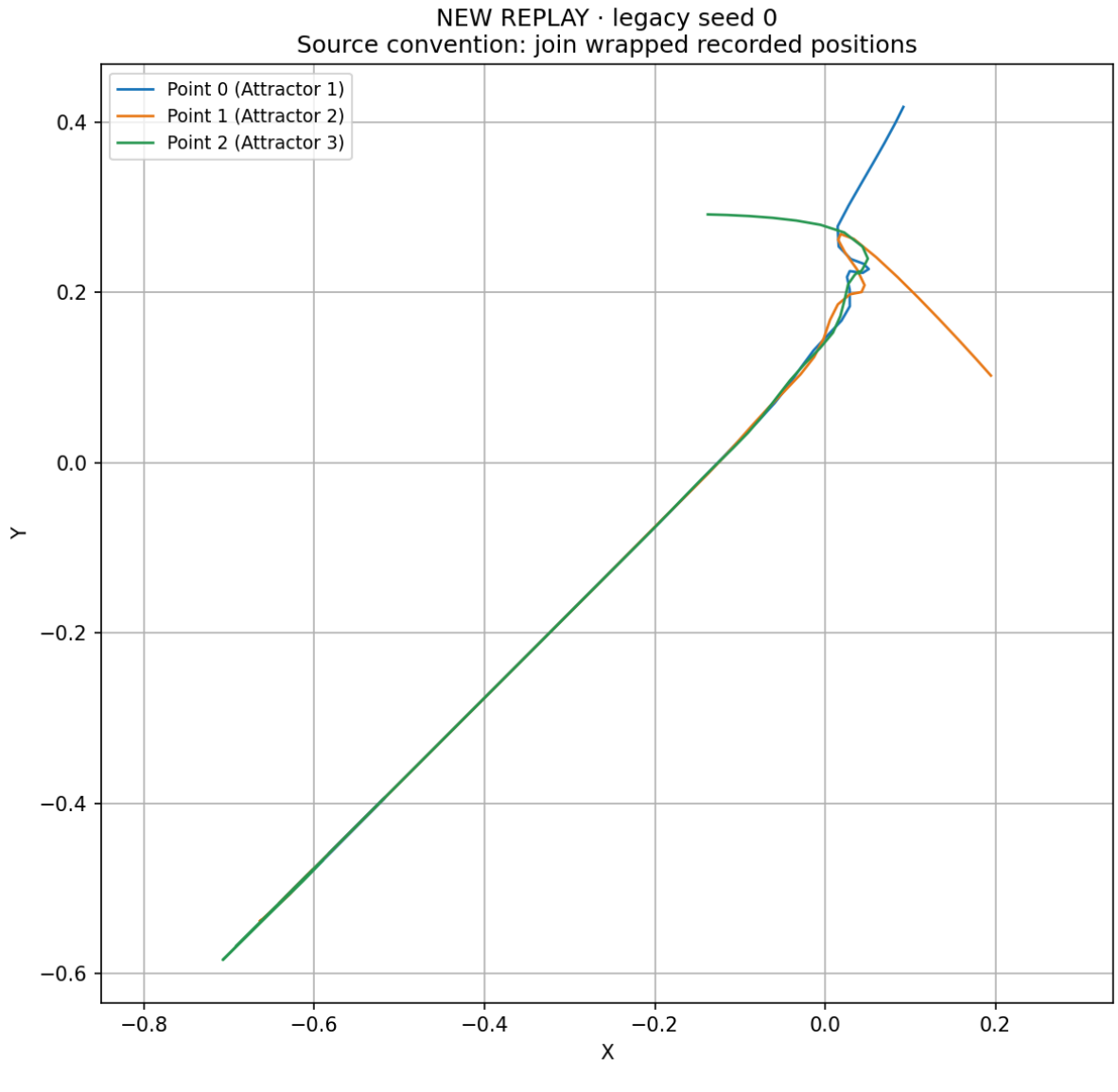


Figure 2.2: REPRODUCED / OBSERVED. Accepted seed-0 S01 paths, using the source's 500 post-sweep samples. X/Y are coordinate values. The new replay label distinguishes these paths from the historical PNG. Shared drawn segments do not establish simultaneous occupancy or common movement direction; no continuous trajectory between samples is supplied. [5]

for this manuscript. Figure 2.3 makes the ordered dependence visible. [5], within_sweep.jsonl lines 1–9. [C23]

For body 0's first update, the accepted record gives approximately

$$F_0 = (-0.570665, -1.433338),$$
$$u_0 = (-0.128400, -0.322501),$$
$$v_0^+ = (-0.114561, -0.252047).$$

The damped result is the next incoming velocity. All three points move in the opening sweep of this reproduction; this observation is not a general theorem excluding small-motion waiting under every initializer. [C24]

2.5.2 Near-contact and subsequent separation

The accepted replay selected the smallest simultaneous periodic separation among the 500 recorded end-of-sweep states and three unordered pairs, with a declared tie rule. It selected bodies 1 and 2 at S_{169}. The inspected events are 502–510, with the preceding state at event 501. The central sweep uses source step 168, $t = 8.4$ and $H(t) \simeq 0.09211406$. [C25]

Table 2.3: A finite reproduced near-contact. Distances are in coordinate units at recorded sweep boundaries, not interpolated continuous minima. Source: replay/observation_summary.json and within_sweep.jsonl events 504,507,510.

State	Completing event	Distance between bodies 1 and 2
S_{168}	504	0.001496078692
S_{169}	507	0.0000442435133
S_{170}	510	0.001583353616

At the central sweep, body 0's interaction points up-right but its outgoing velocity remains approximately $(-0.040387, -0.040422)$. Bodies 1 and 2 have down-left interaction sums while their outgoing velocities remain approximately $(0.008924, 0.008872)$ and $(0.030815, 0.030547)$. Carried velocities explain why the instantaneous interaction direction does not identify the movement direction. The two nearly overlapping markers conceal distinct coordinates and velocities; the next sweep separates them again. [5], events 505–507 and 510. [C26]

This is a measured near-contact in one reproduced chronology. It is not exact coincidence, a proven continuous collision, or a delayed leader–pair route-following episode. The point moving the other way along the same diagonal demonstrates why a completed line picture is insufficient for that interpretation. [C27]

2.5.3 The complete finite record

The accepted record has no position seam crossings in 1,500 point updates and no nonzero wrap shift in 3,000 interaction comparisons. The wrapping operators still execute. The largest native increment is about 0.03456666287 at event 20, body 1 in source step 6. The ordinary and seam-aware paths therefore have the same curves for this initializer. This absence is not a property proved for every execution of S01. [C28]

Figure 2.5 shows repeated approach and separation together with smaller later increments. All final velocities at S_{500} remain nonzero. The accepted record also contains seven sign changes of its sampled H values; no ablation was performed to assign a full episode to H alone. Finite slowing does not establish eventual convergence. [5], observation_summary.json and record_checks.json. [C29]

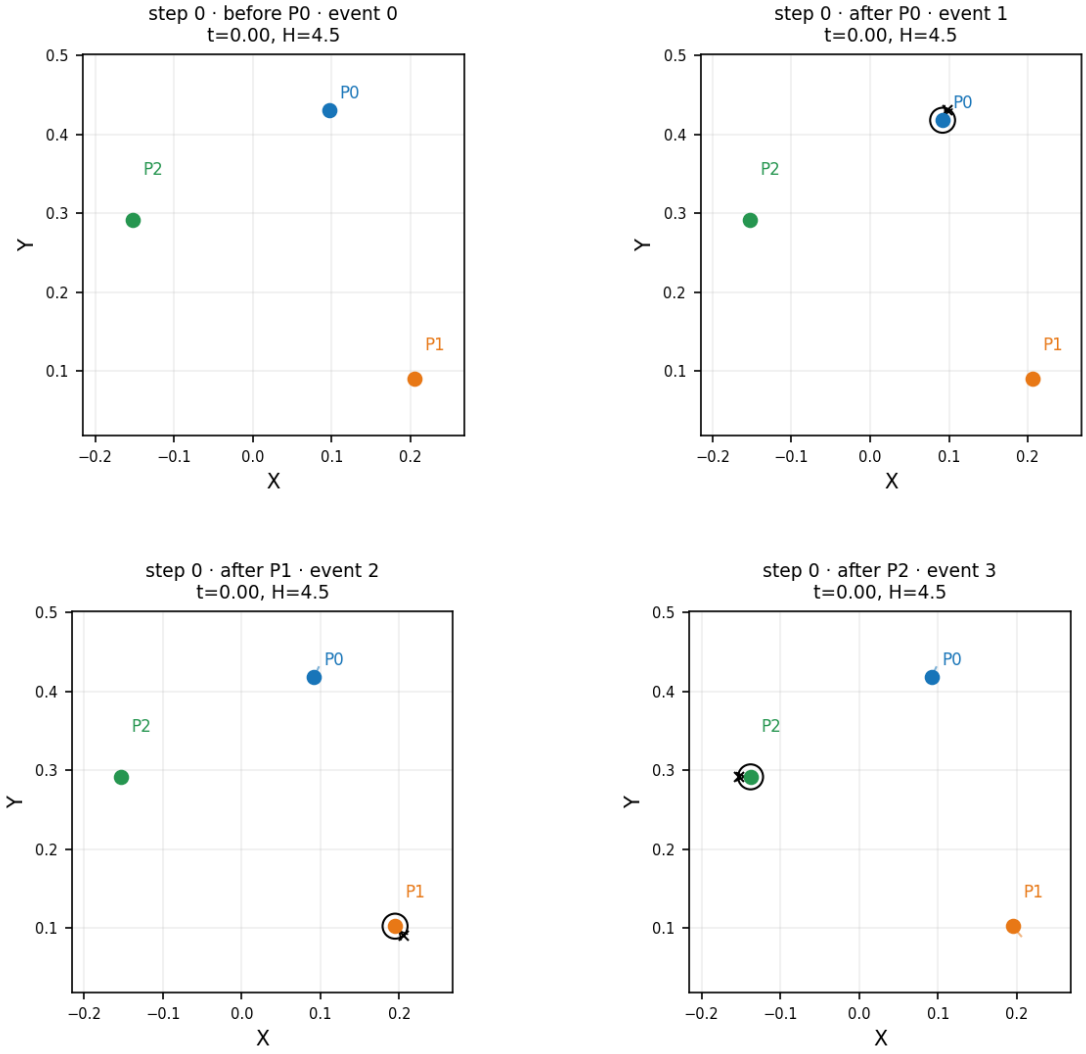


Figure 2.3: REPRODUCED / OBSERVED, editorial crop. Four panels from the first row of the accepted opening sheet, arranged in reading order: before body 0, after body 0, after body 1, after body 2 (events 0–3). The first panel is the captured initializer; the others are recorded states. X/Y axes retain their original ranges. Arrows connect observed endpoints, not simulated intermediate positions. The source sheet also contains the next two sweeps. [5]

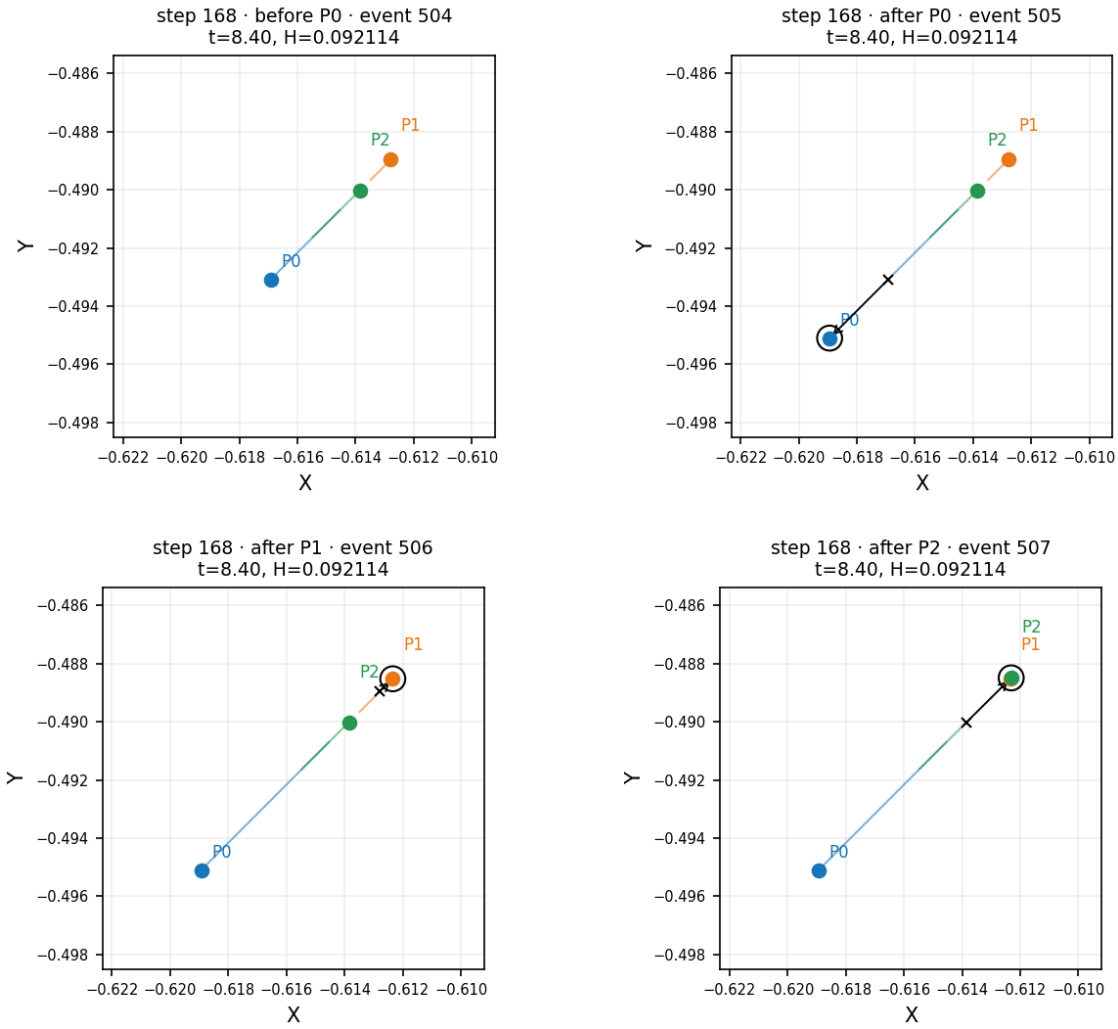


Figure 2.4: REPRODUCED / OBSERVED, editorial crop. Central row of the accepted near-contact sheet, arranged in reading order: event 504 before the sweep, then events 505,506,507. Source step 168 uses t=8.4. The local X/Y zoom makes the final near-overlap visible, while the saved velocities remain distinct. Table 2.3 supplies the subsequent separation; no exact coincidence or interpolated meeting is asserted. [5]

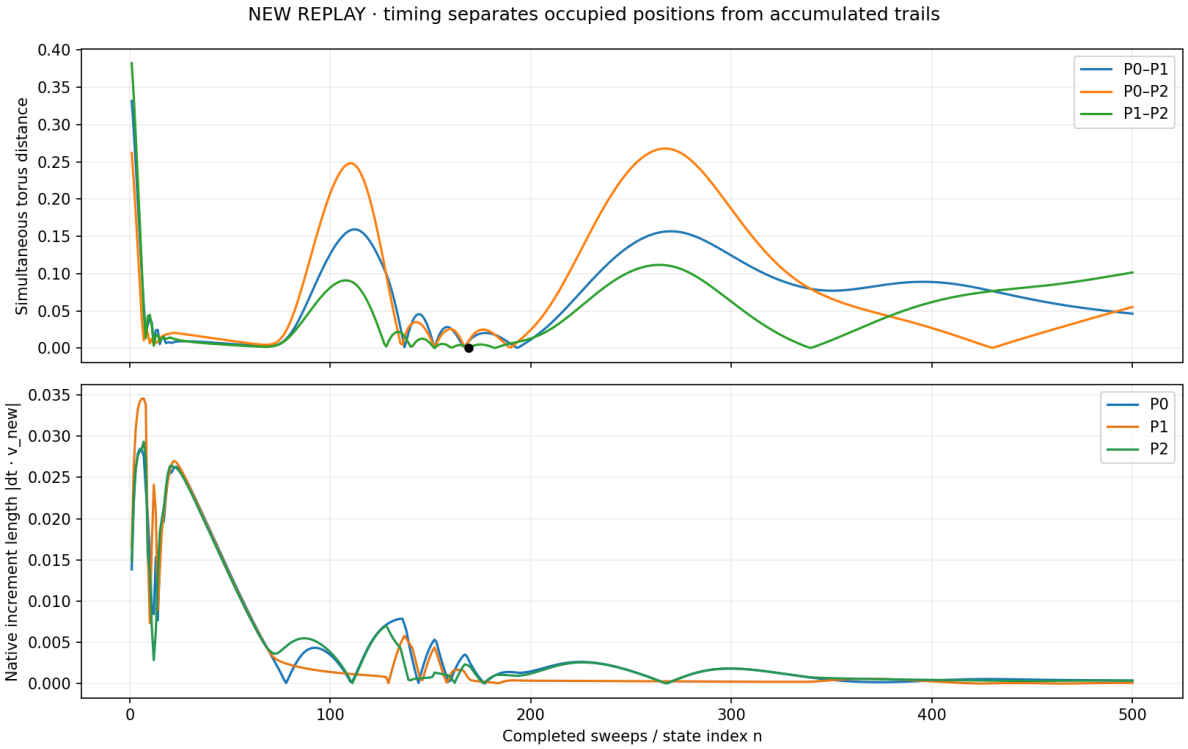


Figure 2.5: DERIVED from the accepted REPRODUCED S01 record. The upper panel measures simultaneous periodic pair separation at sweep boundaries; the lower measures native update lengths. The horizontal axis is completed-sweep chronology, not historical movie time. Repeated approaches and smaller late movement do not establish an asymptotic limit. Reused without recalculation. [5]

Chapter 3

Simulation 02: eight tetrahedral centers

In ordinary language. This program moves eight centers through three coordinate directions. The tetrahedra are fixed shapes drawn around them. Each center takes its turn responding to an average of the currently stored center coordinates. The movie shows the resulting shapes, but the missing numerical history limits what can be measured from it.

3.1 The world and the drawing geometry

The complete 119-line `3Dtriangles.py` initializes an $(8, 3)$ position array uniformly through `np.random.rand(8,3)*10` and an $(8, 3)$ velocity array through `(np.random.rand(8,3)-0.5)*0.2`. Thus positions lie in $[0, 10)^3$ and velocity components in $[-0.1, 0.1)$. No fixed seed is set. Unlike S01, the initial velocities are generally nonzero. There are 48 real-valued carried entries. S02, lines 8–16. [C30]

The relative vertex offsets are

$$\tfrac{1}{2}(1,1,1), \quad \tfrac{1}{2}(-1,-1,1), \quad \tfrac{1}{2}(-1,1,-1), \quad \tfrac{1}{2}(1,-1,-1),$$

with face index triples $(0, 1, 2)$, $(0, 1, 3)$, $(0, 2, 3)$ and $(1, 2, 3)$. Their edge length is $\sqrt{2}$, directly from the coordinate differences. Every displayed tetrahedron retains its size and orientation. Its center translates; its vertices do not have velocities, force laws or collision rules. Face transparency is 0.6. No independent vertex wrapping is implemented. S02, lines 19–26,64–71,99–102. [C31]

3.2 Exact centroid construction

Write the componentwise position and displacement wraps as

$$\mathcal{W}_{10}(z) = z \bmod 10, \qquad \mathcal{D}_{10}(z) = ((z + 5) \bmod 10) - 5. \tag{3.1}$$

These map respectively to $[0, 10)$ and $[-5, 5)$ in each coordinate. At a callback with supplied label f, let x_i, v_i be the incoming state and x_i^+, v_i^+ the resulting state. The source sets

$$t = 0.1f, \qquad h = H(t), \qquad g = \frac{1}{1 + \frac{1}{8}\sum_{j=0}^{7} \|v_j\|_2}. \tag{3.2}$$

H and RPCO are the same formulas as Equations (2.1) and (2.3). GRF accepts positions and time as arguments but reads neither; it uses the global incoming velocity array. It is computed once, not recalculated after each body moves. S02, lines 29–51,74–82. [C32]

For $i = 0, \ldots, 7$ in order,

$$c_i = \frac{1}{8}\left(\sum_{j<i} x_j^+ + \sum_{j\geq i} x_j\right), \tag{3.3}$$

$$d_i = \mathcal{D}_{10}(c_i - x_i), \qquad r_i = \|d_i\|_2, \tag{3.4}$$

$$a_i = 0.1gh\,\frac{d_i}{r_i + 0.01}, \tag{3.5}$$

$$u_i = v_i + a_i, \qquad v_i^+ = \mathcal{R}(u_i), \tag{3.6}$$

$$x_i^+ = \mathcal{W}_{10}(x_i + 0.1v_i^+). \tag{3.7}$$

The symbol a_i denotes the source's velocity contribution, not a new mechanism. The centroid includes the body itself and averages the currently wrapped Cartesian coordinates. Earlier bodies have already moved. The regularized direction has magnitude $r_i/(r_i + 0.01)$, multiplied by the common signed H and GRF factors. H is multiplied exactly once. S02, lines 83–96. [C33]

There is no binary switch in this program and no random draw after initialization. Nevertheless, "attraction to centroid" is a source comment rather than a guarantee of inward velocity. H can be negative, and even a positive inward contribution may leave an outward carried velocity after addition and damping. GRF lies in $(0, 1]$ for finite velocities and decreases as mean incoming speed grows. S02, lines 33–51,83–96; [6]. [C34]

3.3 Clock labels, animation and retained state

The source requests 2,000 animation labels, a 50 ms interval and `blit=True`. It supplies no separate initialization function. It attempts a 60 fps MP4 save and then calls `plt.show()`, also showing after a caught save exception. Arrays are not reset between save and display. All of this is at module level. S02, lines 107–119. [C35]

Table 3.1: Four distinct time-related quantities in S02. The first two affect the numerical rule; the last two control requested presentation.

Quantity	Meaning
$\Delta t = 0.1$	Multiplier in both velocity addition and position advance
$t = 0.1f$	Argument passed to H on each callback; not an accumulator
GUI interval 50 ms	Nominal timer interval, not a measured callback duration
Movie 60 fps	Encoded presentation cadence, independent of model-step units

If labels 0–1999 are processed once, H's argument ranges from 0 to 199.9, while 2,000 multipliers of 0.1 account for 200 model-step units. That conditional count does not establish the historical number of state-changing callbacks. The accepted local-library inspection (Matplotlib 3.10.9) found initialization, save, interactive-start and repeat paths that can invoke a callback with its first label. With a mutating update function, an initialization draw is a numerical update, not necessarily a harmless rendering of the initializer. Historical dependency versions and backend events are unrecovered. [6], TIME_AND_MOVEMENT and saved library excerpts. [C36]

Repeated labels can apply the same H value to changed arrays. A label reset can renew the large starting gain while retaining positions and velocities. This is a source/library possibility for S02; its movie does not certify that exact callback chronology. S03's surviving logs will supply direct evidence of repeated labels for that distinct program. [C37]

3.4 Wrapping can affect both the image and the interaction

From the position line, there is an integer coordinate vector k_i such that

$$x_i^+ - x_i = 0.1v_i^+ - 10k_i. \tag{3.8}$$

The native increment $0.1v_i^+$ can therefore be small while a stored coordinate difference is almost ten. The displayed shape moves with the wrapped center; the movie does not draw a periodic seam connection or interpolate a path across it. S02, lines 95–101. [C38]

There is a second, distinct effect. Because c_i is an arithmetic mean of wrapped coordinates, a body's positive crossing contributes $-10/8 = -1.25$ to that coordinate of the group average, in addition to ordinary motion. Later bodies in the same sweep see the changed average. Wrapping $c_i - x_i$ afterward does not turn the preceding mean into an intrinsic mean on a torus. The chosen displacement itself also has a branch at a half-period. These are deductions from the implemented operations, not proposed replacements. S02, lines 85–96; [6]. [C39]

In ordinary language. A boundary relocation can be a drawing jump and still matter to later motion. This program averages the displayed coordinate representatives before wrapping the direction. We must preserve that choice to explain the historical construction.

3.5 The original movie and what it shows

The attributed movie `tetrahedral_attractors_animation.mp4` contains 2,000 decoded frames at 60 fps, 1000 by 800 pixels, duration $33\frac{1}{3}$ seconds. Its name, title, geometry, axes and nominal encoding agree with the source. The original lies in part_1 while the source lies in part_2; the output's relative working-directory path makes that directory difference insufficient to reject attribution. No source/run receipt or ordered numerical history was recovered. [6], VISUAL_EVIDENCE. [C40]

Figure 3.1 samples the full duration. Shapes become closer in projection and later arrangements change much less visibly. Several shapes remain separated in the final image. These observations establish neither every three-dimensional pair distance nor exact rest. The accepted study decoded all frames but visually reviewed selected sheets and full-resolution stills; it did not claim to have manually inspected every frame. [C41]

The consecutive window in Figure 3.2 locates a conspicuous screen relocation between frames 13 and 14. It is consistent with the source wrap operation, but the missing positions and velocities prevent certification of the crossing axis and native displacement. No value from S03's CSVs is substituted for this missing S02 history. [C42]

Color is another observation variable. The source uses `viridis((i+frame)%8/8)`, so an individual body's color changes even when its location changes little. The first eight frames (Figure 3.3) illustrate why color alone is not a stable identity label. Projection and transparency can overlap tetrahedra separated in depth; their silhouettes may look triangular despite four-face source geometry. There are no accumulated trajectory lines in this movie, unlike S01's static plot. S02, lines 64–71,99–102. [C43]

The source supports coupling through the centroid, carried velocity and shared speed feedback. The inspected movie supports visible motion, one localized relocation and slower late changes. It does not establish a leader, route-following pair, collision, asymptotic equilibrium or physical gravitational interpretation. These limits concern available evidence, not a theorem that the program can never display another collective event. [C44]

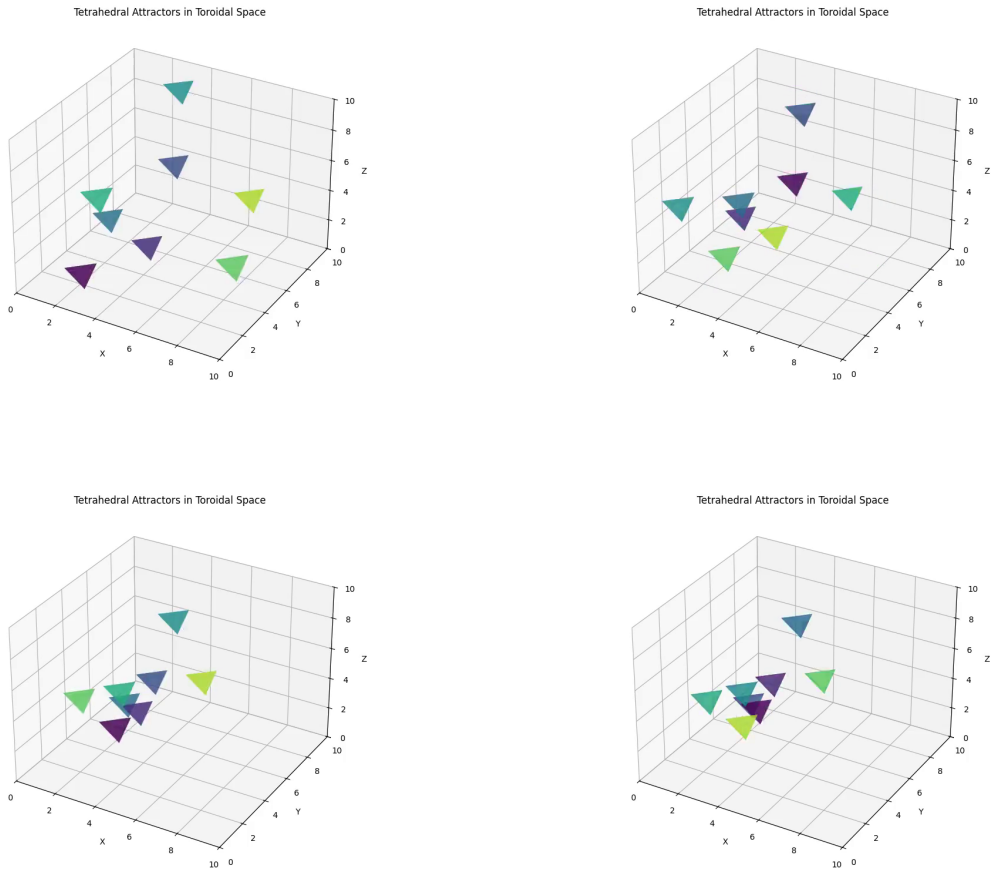


Figure 3.1: ORIGINAL movie content / ATTRIBUTED, decoded stills. S02 frames 0,30,600,1999 in reading order (0,0.5,10,33.3167 movie seconds). The XYZ axes cover 0–10 in source coordinates. Projection shows changing arrangements and smaller late screen motion, not measured 3D speeds or convergence. Decoded in the accepted Construction 02 study; no new state was generated. [6]

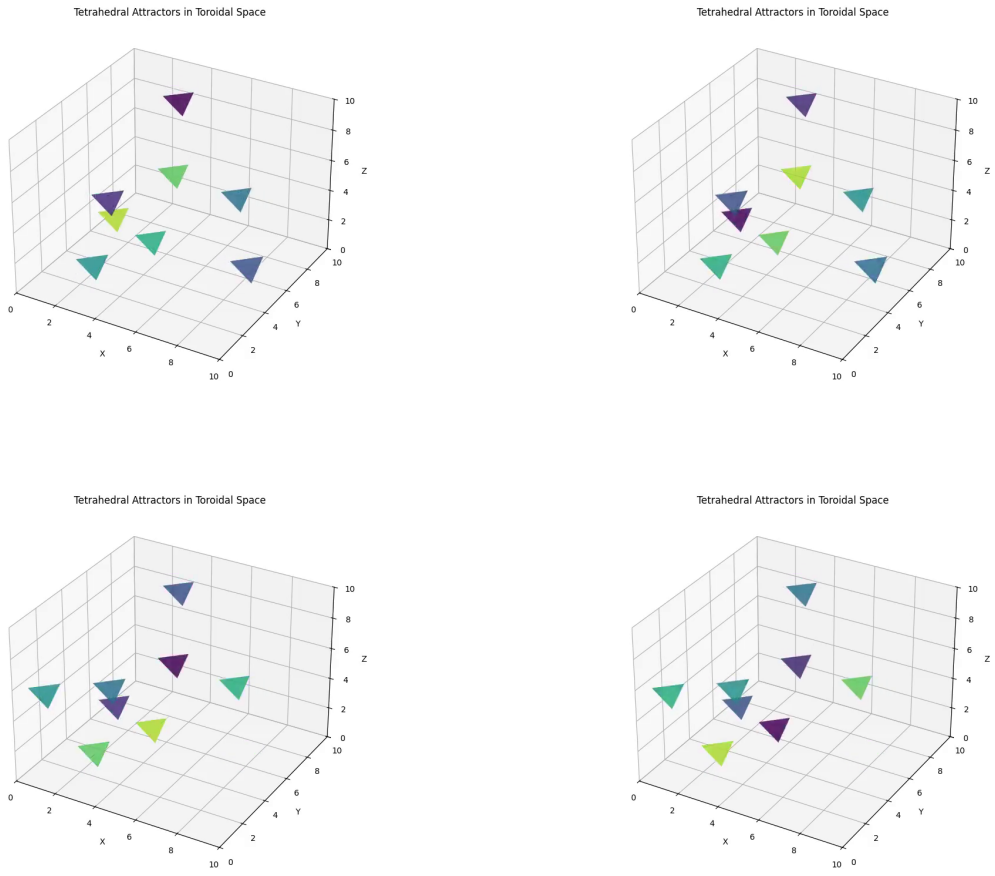


Figure 3.2: ORIGINAL movie content / ATTRIBUTED. Consecutive S02 frames 12–15 in reading order (0.2000–0.2500 movie seconds). The conspicuous relocation occurs between 13 and 14 and is consistent with wrapping. Missing numerical states prevent a certified crossing axis or native displacement. Color also advances between frames and is not a stable body label. [6]

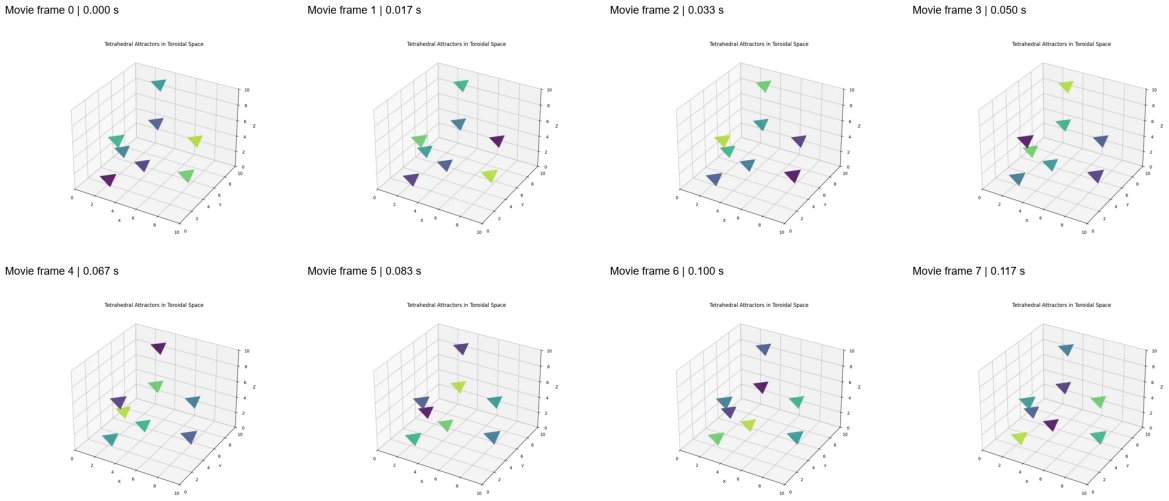


Figure 3.3: DERIVED inspection sheet of ORIGINAL / ATTRIBUTED movie frames 0–7, read left to right then down. Frame labels and movie seconds were added by Construction 02. Color cycling follows body index plus frame even when position changes little. These colors do not measure velocity or force. The fixed view can overlap separate 3D objects. [6]

Chapter 4

Simulation 03: switching and recorded history

In ordinary language. Each center now carries a switch as well as its position and velocity. The switch reverses the new interaction contribution, not the motion already being carried. The saved logs reveal the difference. They also show that the program can restart its labelled clock without restarting its state. This chapter follows those records before interpreting the movie.

4.1 Construction and added state

The 167-line `plot_1.py` retains eight 3D centers, period 10, $\Delta t = 0.1$, the same tetrahedral offsets, random initial coordinates and random velocity components. It adds eight binary modes b_i, all initialized to zero. Its numerical carried state therefore has 48 real entries and eight mode bits, together with the random stream and externally supplied frame input. S03, lines 10–28,152; [7]. [C45]

The common H, componentwise RPCO and mean-speed GRF function bodies match S02 after removing docstrings, as recorded by the accepted AST check. At each callback, GRF and H are computed once, then the incoming positions are copied for a diagnostic. Bodies update in order 0–7, with the same mixed-state arithmetic centroid and wrapped displacement. S03, lines 36–54,102–113. [C46]

The source is not merely S02 plus a color change. It also changes nominal frame count to 200, requests `blit=False`, saves at 20 fps, computes a kinetic diagnostic, prints a jump diagnostic and appends numerical records on every callback. These changes affect the available chronology and presentation even though they do not all feed into the movement rule. S03, lines 12,57–78,129–167. [C47]

4.2 Exact switch and update equations

Use $\mathcal{W}_{10}$, $\mathcal{D}_{10}$ and $\mathcal{R}$ from Equations (3.1) and (2.3). At callback q, with incoming velocities v_j and supplied label f_q, set

$$g_q = \frac{1}{1 + \frac{1}{8}\sum_j \|v_j\|_2}, \qquad h_q = H(0.1 f_q). \tag{4.1}$$

For each body i, calculate $c_{q,i}, d_{q,i}, r_{q,i}$ using Equations (3.3)–(3.4). Only then evaluate the gate. If $g_q > 0.5$, draw $U_{q,i}$ using `np.random.rand()`; otherwise no draw is consumed. Define

$$F_{q,i} = \begin{cases} 1, & g_q > 0.5 \text{ and the consumed draw } U_{q,i} < 0.01, \\ 0, & \text{otherwise,} \end{cases} \qquad b_i^+ = b_i \text{ XOR } F_{q,i}. \tag{4.2}$$

The update is

$$a_{q,i} = 0.1 g_q h_q (1 - 2b_i^+) \frac{d_{q,i}}{r_{q,i} + 0.01}, \tag{4.3}$$

$$u_{q,i} = v_i + a_{q,i}, \qquad v_i^+ = \mathcal{R}(u_{q,i}), \tag{4.4}$$

$$x_i^+ = \mathcal{W}_{10}(x_i + 0.1 v_i^+). \tag{4.5}$$

The sign is the *post-toggle* mode, the one recorded in the current block. H is multiplied once, and GRF remains the incoming common value throughout the sweep. S03, lines 102–126. [C48]

Conditional on the intended uniform generator and an open gate, the toggle probability is 1% per body per callback. No random values or RNG state are logged. When H is positive, mode 0 contributes toward the chosen wrapped centroid displacement and mode 1 away from it. When H is negative those roles reverse. Neither mode alone specifies the outgoing velocity direction. [C49]

4.3 Recording and callback chronology

At startup the source writes an eleven-column header to a timestamped CSV. After each complete callback it appends eight rows in body order:

```
Step,Tetrahedron,X,Y,Z,VX,VY,VZ,Energy,GRF,Rule_State.
```

Positions, velocities and modes are outgoing values; Energy is the whole-system post-sweep value repeated in each row; GRF is incoming. There is no saved detector flag, H column, within-sweep snapshot or per-row wall-clock time. S03, lines 74–78,129,144–147. [C50]

Table 4.1: Original log structure, preserved in physical row order. The repeated-key count is the number of rows beyond the first occurrence of each (Step,Tetrahedron) key. Accepted source: S03 ORIGINAL_LOG_ANALYSIS and analysis/log_summary.json.

Measure	Log A	Log B
Data rows	8,736	3,520
Eight-body callback blocks	1,092	440
Distinct frame labels	200	200
Label-zero blocks	12	6
Repeated-key rows	7,136	1,920
Partial or corrupt blocks	0	0

Log A is `PLOT_1_tetrahedral_states_20250712_015421.csv`; log B is `tetrahedral_states_20250712_015647.csv`. They contain finite values, correctly ordered body IDs and shared Step/Energy/GRF within each block. All positions lie in $[0, 10)$ and all modes are binary. They are strongly attributed to this recurrence by schema and numerical agreement; the original seed and pre-log state are missing. [C51]

Each label-zero block is a distinct recorded state. Sorting by Step and discarding duplicates would destroy the chronology in Table 4.2. Counting 0.1 per recorded callback gives

Table 4.2: Exact label chronology. A dash denotes no further record, not an assumed reset or simulated continuation.

Callback range	A labels	B labels
0	0	0
1–200	0–199	0–199
201	0	0
202–401	0–199	0–199
402	0	0
403–602	0–199	0–36 through q439 only
603	0	—
604–803	0–199	—
804	0	—
805–1004	0–199	—
1005	0	—
1006–1091	0–85	—

109.2 and 44.0 model-step units; the first-to-last stored-state spans are 109.1 and 43.9. Neither count is the variable supplied to H: its argument repeatedly returns to zero. [7], ORIGINAL_LOG_ANALYSIS. [C52]

The accepted inspection of local Matplotlib 3.10.9 identifies initialization, save, interactive start and repeated playback as possible sources of extra first-label calls. The observed singleton zeros and complete passes are consistent with those paths. However, the historic library, backend and GUI event sequence are missing. We preserve recorded labels without assigning an invented save or window event to every segment. [C53]

4.4 What the accepted numerical checks cover

The accepted primary checker read every row. It reconciled the kinetic quantity on all 1,532 blocks, and position and incoming-GRF relations on every consecutive block: 8,728 body transitions in A and 3,512 in B. Callback zero lacks a recorded predecessor, so its incoming update cannot be certified. [C54]

The full velocity equation was checked at a declared union of sample callbacks: the first five transitions, final five, maximum native increment, strongest consecutive-vector turn, closest pair, first label discontinuity, first decreasing-label reset, first switch and first seam. Deduplicating that *selection list*, not the data, gave 16 sampled transitions in A and 15 in B. The checker substituted the observed current prefix and preceding suffix into the centroid; it did not propagate calculated states. An independent scalar implementation checked the 20 original R0 sample transitions and all energy/position/GRF identities. [7], equation-residual CSVs and independent_checks.json. [C55]

All declared checks passed an absolute tolerance of 10^{-12}. Across the methods, reported maximum position residual was 1.78×10^{-15} and sampled velocity residual 3.61×10^{-16}. Such agreement supports source attribution and the interpretation of columns. It is not an exhaustive velocity audit, recovery of random draws or bit-identical reproduction of the unknown historical execution. These scientific checkers were not rerun for Volume I. [C56]

4.5 A sign change before movement reverses

All recorded callbacks have GRF above 0.5. Its range is approximately 0.633309–0.987992 in A and 0.641949–0.987362 in B. Between successive recorded blocks there are 79 toggles in A and 33

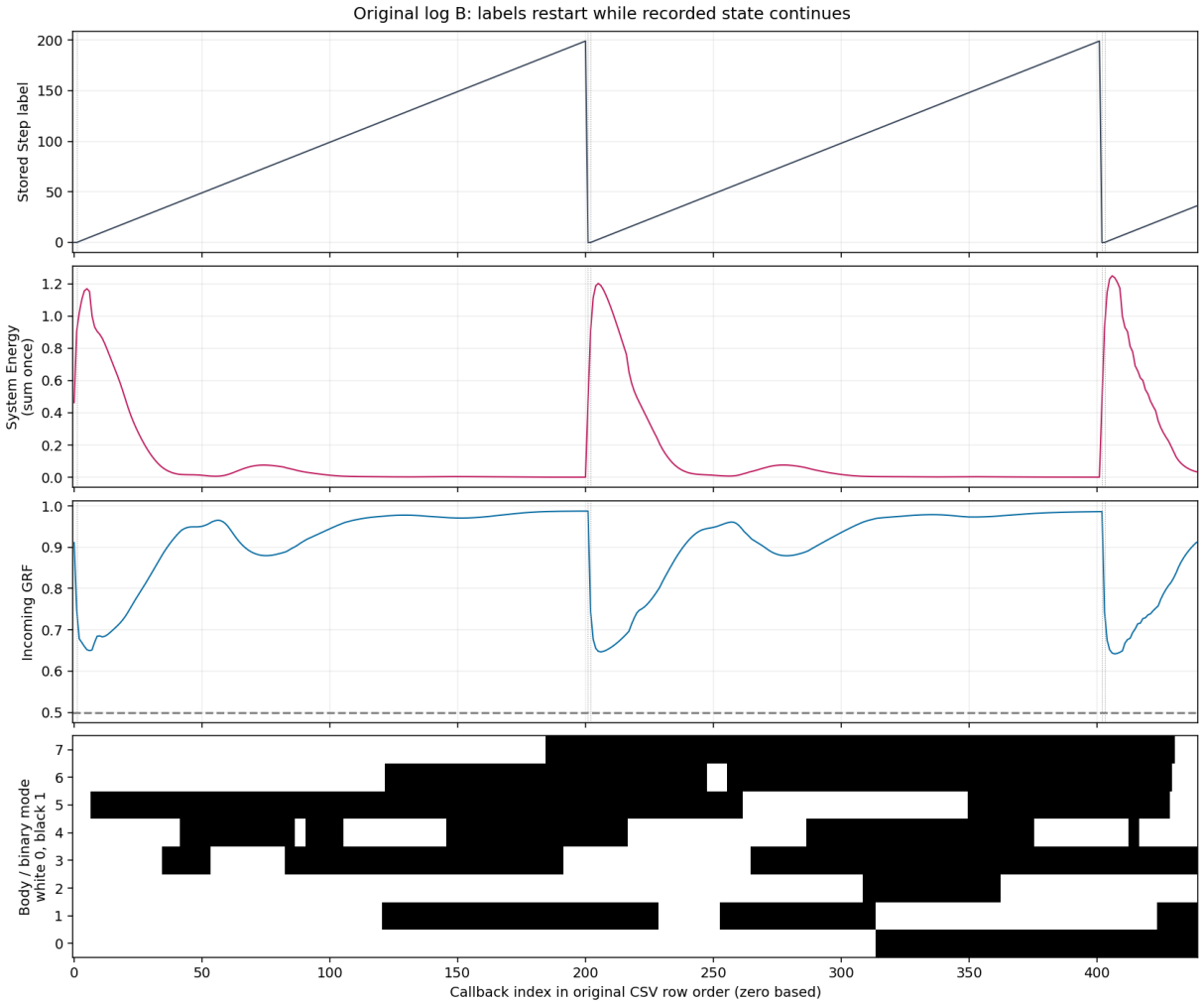


Figure 4.1: DERIVED from ORIGINAL log B, reused from Study 03. Horizontal axis: zero-based callback q in physical CSV order. Panels show Step, global K counted once, incoming GRF and binary modes. Vertical dotted lines mark label discontinuities. State persists across returns to label 0; 440 callbacks are not 200 unique states. No GUI event cause is assigned to each restart. [7]

in B. All are gate-eligible, but the unrecorded draws cannot be reconstructed or used to certify a distribution. [C57]

The shorter log's first toggle is body 5 at $q = 7$, Step 6, CSV line 63. It changes mode from 0 to 1 while H remains positive. Table 4.3 follows the recorded response. The radial component is the dot product of outgoing velocity with the unit wrapped centroid displacement evaluated at that body's update, using the observed mixed-state positions. Positive means along that instantaneous direction. [C58]

Table 4.3: Original log B, body 5. A response-sign change at q7 precedes an outward radial velocity at q9. Source: original CSV lines shown; accepted analysis/episode_records.json. Rounded values, no new update.

Callback	CSV line	Mode	Speed	Radial outgoing velocity
6	55	0	0.570163	$+0.565607$
7	63	1	0.291844	$+0.284300$
8	71	1	0.087759	$+0.062268$
9	79	1	0.141476	-0.131445

The new opposing contribution first reduces an inward carried velocity. The next record still moves inward; two callbacks after the toggle, its radial outgoing component is negative. This is a specific delayed direction change, not a universal two-callback law. A's first toggle, body 4 at $q = 12$, has an analogous ordering through $q = 14$. [C59]

Across all observed toggles, none has a negative dot product between the immediately preceding and outgoing velocity vectors. There are 17 such obtuse consecutive turns in A and 6 in B at other transitions. The strongest examples are A body 4 at $q = 1005$ (line 8046, cosine about -0.977692) and B body 5 at $q = 402$ (line 3223, about -0.973052). Both retain mode 1 and occur when the frame label returns to zero. A switch, a velocity reversal and a clock reset are distinguishable events. [7], switch-event tables and log_summary.json. [C60]

4.6 Energy accounting and renewed clock drive

The function `compute_energy` ignores its positions argument. It returns

$$K^{+} = \frac{1}{2}\sum_{i=0}^{7} \|v_i^{+}\|_2^2 + 0. \tag{4.6}$$

The zero is a literal potential-energy placeholder. No mass parameter, position potential, conservation rule or energy feedback is implemented. The same global number is written eight times per callback; summing the column over a block would count it eight times. S03, lines 57–61,129,147. [C61]

At the first $199 \to 0$ label reset, $q = 200 \to 201$, A's K changes from 0.000922772 to 0.479749699, and B's from 0.000721817 to 0.473368802. The source changes the prescribed gain from $H(19.9) \simeq 0.00261412235$ to $H(0) = 4.5$ while retaining state. Another zero follows, supplying another strong-drive update. Figure 4.1 shows this chronology for B. These are recorded drive/damping effects, not spontaneous creation of a physically conserved energy. [C62]

For either centroid update, let a_i be the source contribution, $u_i = v_i + a_i$, and $v_i^{+} = \mathcal{R}(u_i)$.

Directly expand the squared norm and then subtract the damping loss:

$$\tfrac{1}{2}\|u_i\|^2 - \tfrac{1}{2}\|v_i\|^2 = v_i \cdot a_i + \tfrac{1}{2}\|a_i\|^2, \tag{4.7}$$

$$L_i = \tfrac{1}{2}\big(\|u_i\|^2 - \|\mathcal{R}(u_i)\|^2\big) \geq 0, \tag{4.8}$$

$$K^+ - K = \sum_i v_i \cdot a_i + \frac{1}{2}\sum_i \|a_i\|^2 - \sum_i L_i. \tag{4.9}$$

The inequality follows componentwise from the denominator in RPCO, which is at least one. This is the accepted source-derived bookkeeping identity, not an introduced potential or a new dynamical law. The accepted 31-transition sample reports a balance residual no greater than 6.67×10^{-16}, alongside the independent velocity-equation checks. [7], TIME_SWITCH_AND_MOVEMENT and equation-residual tables. [C63]

In ordinary language. The diagnostic can rise because the added interaction changes velocity before damping reduces it. Restarting the H clock restores a strong input. The source contains no statement that this kinetic number must remain constant.

4.7 Native movement, coordinate seams and the detector defect

The position equation separates native increment $\delta x_i = 0.1 v_i^+$ from stored coordinate difference $x_i^+ - x_i$. Equation (3.8) explains their integer-period offset. In the accepted logs, all checkable native increments are below 0.062. In contrast, the largest stored jumps are almost ten. Table 4.4 preserves both measurements and their anchors. [C64]

Table 4.4: Original recorded movements, not image-derived estimates. Anchors use physical CSV lines. Accepted source: S03 log summary and wrap-event tables.

Measure	A	B
Maximum native increment	0.061531942; q205, body 2, Step 3, line 1644	0.060380826; q205, body 5, Step 3, line 1647
Maximum stored jump	9.997419049; q1038, body 4, Step 32, line 8310	9.993505332; q430, body 5, Step 27, line 3447
Native increment at that jump	0.012148730	0.015841001
Actual seam events	22 body / 22 coordinate events	7 body / 7 coordinate events

The source's `detect_teleport` instead computes

$$T_i = \|\mathcal{W}_{10}(x_i^+ - x_i)\|_2 \tag{4.10}$$

and prints an event if $T_i > 1$. This unsigned modulo is neither a minimal periodic distance nor the native increment. A tiny negative coordinate difference becomes almost ten. The diagnostic does not change state, and its flags are not exported in the CSV. S03, lines 64–71,132–134. [C65]

For example, A at q1, body 0, line 10 has coordinate difference

$$\Delta x = (0.040328749, -0.024777022, 0.022525770)$$

and native length 0.052418691, with no seam. The source diagnostic gives 9.975329933 and would flag it. The accepted analysis reports 8,151 would-flag body events in A and 3,129 in B, including 8,130 and 3,122 events without a seam. It misses one of A's 22 actual seams. These are recomputed diagnostics from original records; the original console transcript is missing. [C66]

The first B seam is q2, body 2, Step 1, line 20. Z changes from 0.004472603 to 9.965972877 while the native length is 0.054654713. The corresponding stored jump is 9.961575809. A's earliest

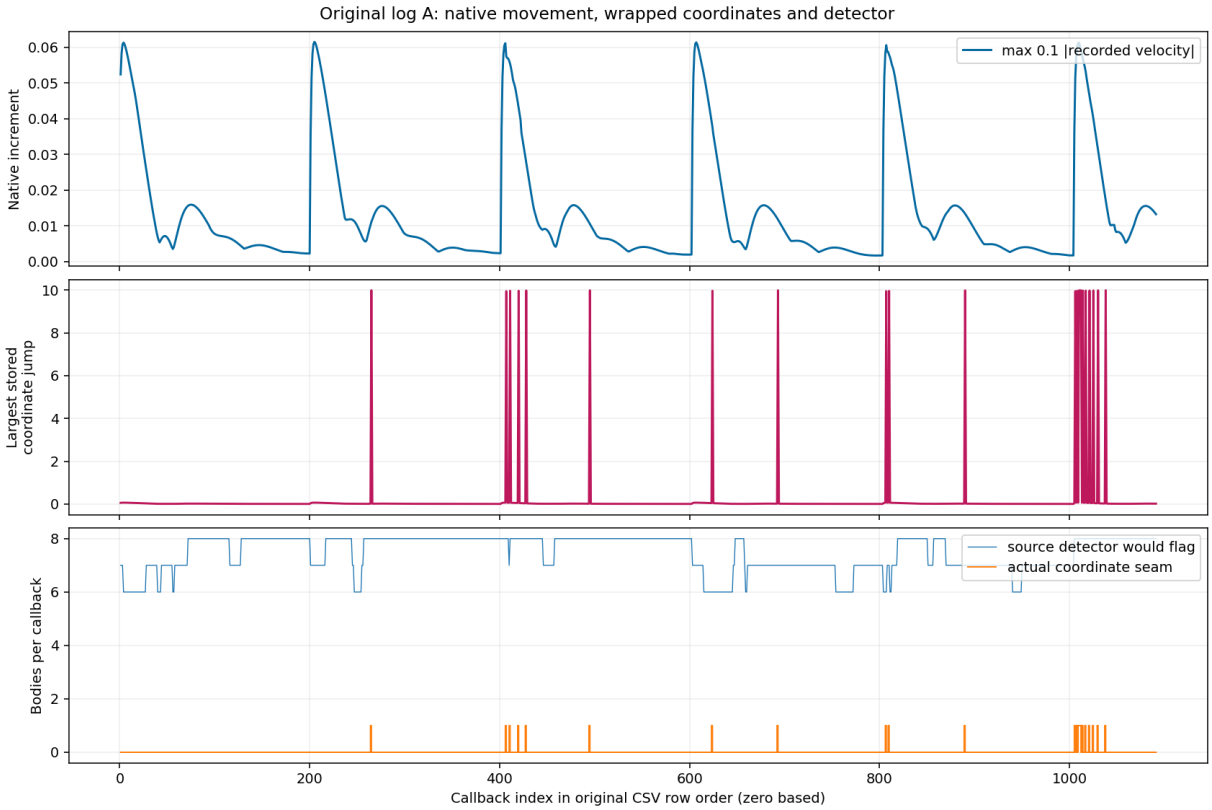


Figure 4.2: DERIVED from ORIGINAL log A, reused from Study 03. Callback order compares native increments with stored-coordinate jumps and source-detector would-flags with actual seams. The 22 seam body events differ from 8151 detector flags. These flags are recomputed diagnostics, not recovered console output or physical teleportation. Values and anchors are retained in the accepted event tables. [7]

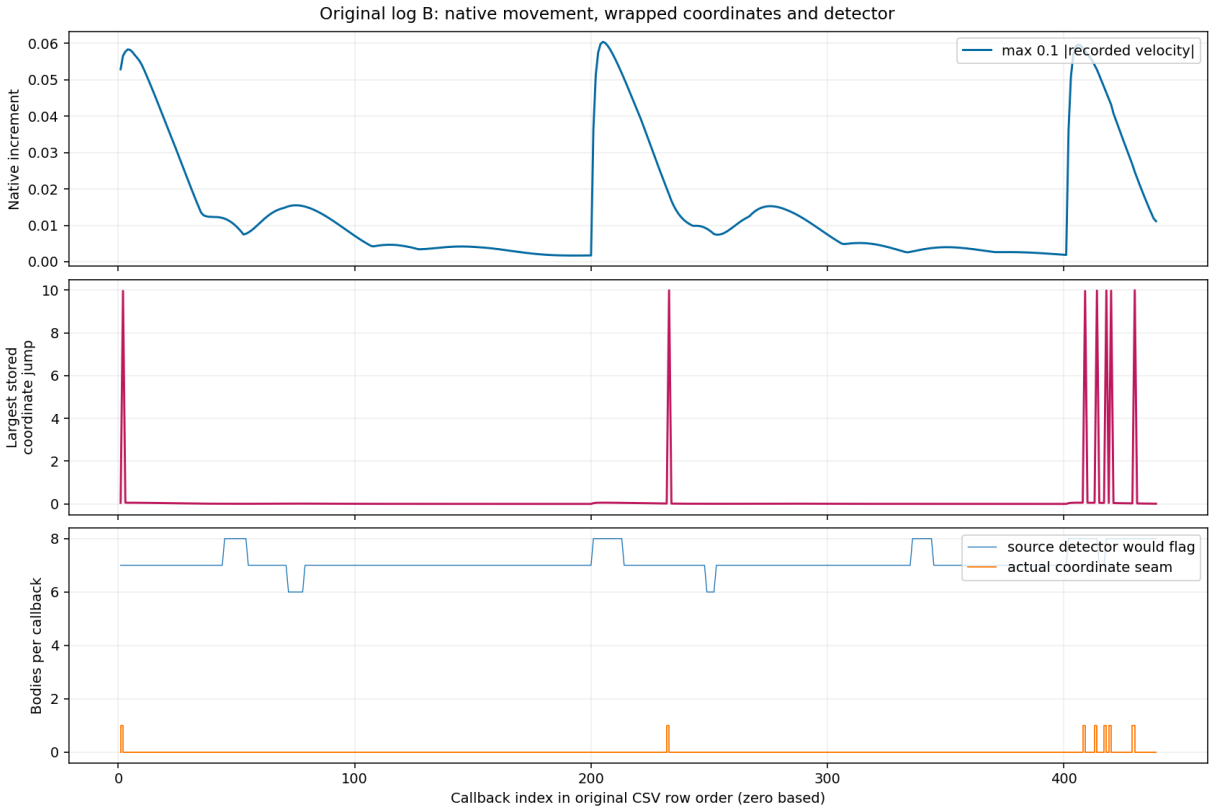


Figure 4.3: DERIVED from ORIGINAL log B, reused from Study 03. Callback order preserves repeated labels. Seven actual seam body events coexist with 3129 detector would-flags; small negative coordinate differences trigger the unsigned-modulo defect. Largest checked native increment is 0.060380826 at q205/body 5. The figure does not equate a near-ten coordinate jump with travel of that length. [7]

seam occurs at q265, body 3, Step 63, line 2125: Z changes from 0.001799251 to 9.998477914 with native length 0.005387221. Neither event is a measured native movement of approximately ten units. [C67]

4.8 Body 7 crosses back and forth without a mode change

In B, body 7 retains mode 1 throughout q400–420. Its X coordinate crosses the seam at q409,414,418, corresponding to lines 3281,3321,3353. The native lengths are approximately 0.055897106, 0.044266228 and 0.037032498. Figure 4.4 separates the coordinate jump, VX and mode. The next mode toggle is q431, outside this displayed window. [C68]

At q409 the center crosses from small positive X to a value near ten. The subsequent wrapped centroid X displacement changes from about $+4.92$ to -3.80. VX then changes from negative to positive, and the center crosses back at q414. The next displacement becomes about $+4.93$ and VX turns negative again before q418. These are accepted measurements of the mixed-state centroid and velocities, retained in analysis/episode_records.json. The arithmetic-mean dependence in Equation (3.3) explains how a coordinate crossing can change the subsequent target. No wall, collision or added teleport mechanism is present. [C69]

The observation supports real sign changes in a velocity component and small native movements through a coordinate seam. It is not a proof of a periodic orbit, nor an isolated causal experiment holding every other body fixed. Other centers continue moving and contribute to the centroid. [C70]

4.9 Pair relations and collective interpretations

The accepted scan of all 28 unordered pairs at stored callbacks found A's closest pair at q602, Step 199: bodies 2 and 7, lines 4820/4825, distance 0.111912869. Their velocity cosine is 0.959590042 and relative speed 0.006143850. That is a near, similarly directed slow pair at a specified instant. B's closest pair occurs at q118, Step 117: bodies 1 and 4, lines 947/950, distance 0.813987821, velocity cosine -0.568814101. Its nearest pair is not aligned at that instant. [7], pair-minima tables. [C71]

A narrow descriptive screen in the accepted study used separation below 0.1, velocity cosine above 0.9 and nonzero speeds above 10^{-6}. Neither log met its distance threshold. This is not a physical threshold, a path-inheritance test or a rejection of approximate collective following. No route-delay criterion was imposed on these records. [C72]

4.10 The original movie and the candidate correspondence

The original `tetrahedral_attractors_logic.mp4` has 200 decoded frames at 20 fps, 1000 by 800 pixels and a duration of 10 seconds. The title, filename, eight tetrahedra, axis range and viridis/Reds palettes match S03's declarations. Palette family indicates mode; within either family its phase changes with $(i+f)$ mod 8. Pale red faces can be difficult to see. No numeric energy or detector-event overlay is drawn. S03, lines 81–97,137–141,158–161. [C73]

The accepted inspection decoded every frame and visually reviewed selected consecutive sheets, an overview, four full-resolution stills and a comparison panel. A bounded attribution test rendered *already-recorded* states, never a calculated trajectory. It compared movie frame 0 with all 18 recorded label-zero states and then tested frames 0,1,6,34,82,120,184,199 against B callback $q = f + 1$. The fixed geometry, view and palette were not optimized. [C74]

At movie frame 0, Bq1 gives colored-pixel mask IoU 0.890282 and mean absolute RGB error

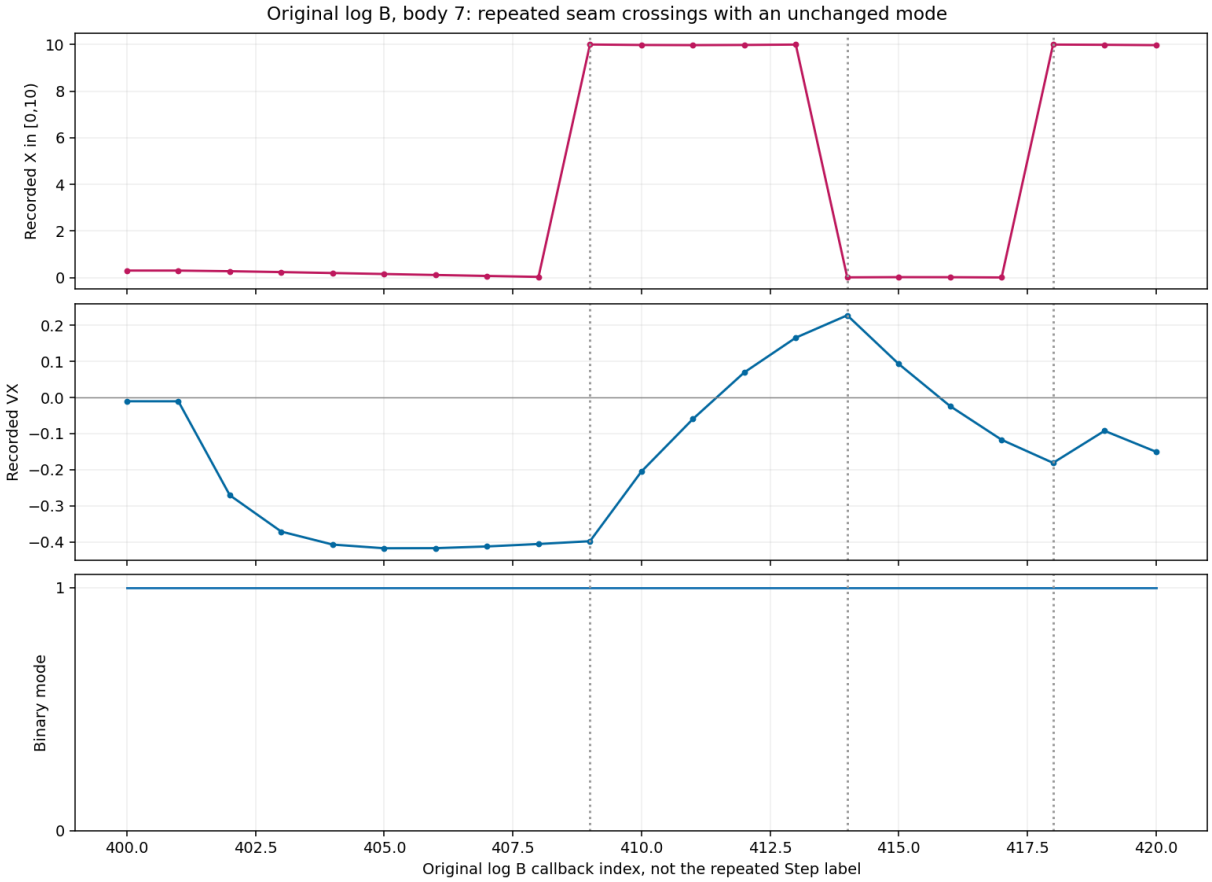


Figure 4.4: DERIVED from ORIGINAL log B, body 7, q400–420. Panels show stored X, VX and mode. Seams at q409,414,418 correspond to CSV lines 3281,3321,3353; the mode remains 1 throughout this window. Coordinate jumps and actual VX sign changes are distinguishable. The later toggle at q431 is excluded. This finite episode is not a periodic-orbit proof. [7]

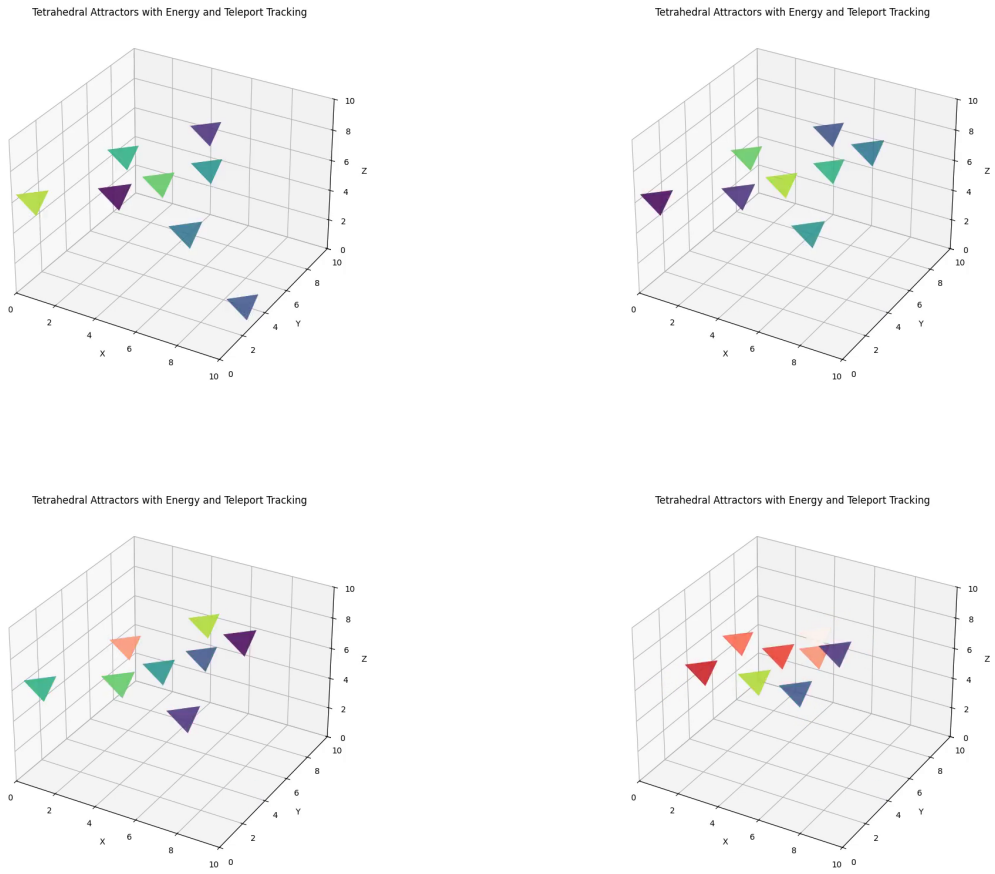


Figure 4.5: ORIGINAL movie content / ATTRIBUTED. S03 frames 0,1,6,199 in reading order (0,0.05,0.30,9.95 movie seconds). The first pair shows an opening screen relocation; later views include red mode palettes. Pale faces can look absent. The title mentions energy/teleport tracking, but no numeric overlay is drawn. Exact original movie/log provenance remains UNLINKED. [7]

5.516925 on the 0–255 scale. Bq0 gives 0.859051 and 13.001680. Across the eight fixed candidate pairs, IoU ranges from 0.856495 to 0.890282 and RGB error from 5.516925 to 7.086700. These descriptive image metrics support a compelling content correspondence, not a calibrated likelihood. Lossy encoding, transparency and rendering versions affect the scores. Figure 4.6 selects two of the already-tested comparisons. [C75]

The formal original-run join remains UNLINKED. No export receipt binds a source hash, RNG state, log and movie. The tested correspondence does not certify all 200 frames or a unique historical execution. Conditionally, the opening movie relocation matches Bq1–2's small Z seam crossing, and frame 6 matches Bq7's first mode toggle. No movie observation silently inherits a CSV measurement as established ground truth. [C76]

4.11 Historical charts and mixed horizontal axes

Two historical body-7 PNGs are strongly attributed to postprocessing of B. The position chart's trace shapes and event markers match the shorter log, but no generating chart script was recovered. S03 itself contains no PNG export. Figure 4.7 preserves the historical chart; Figure 4.8 reuses the accepted diagnostic demonstrating its mixed horizontal coordinates. [C77]

The traces use repeated Step labels 0–199 while markers at 409,414,418 correspond to callback indices of body 7's actual seam crossings. Connecting physical rows on a repeated Step axis folds passes together and joins label 199 back to label 0. The event markers then lie beyond the trace range. This explains a misleading layout without altering historical bytes or claiming exact recovery of the lost plotting script. [C78]

The old GRF/energy chart (Figure 4.9) has the same repeated-label issue. Its Energy comes from the global K repeated on body 7's rows; the title does not make it body 7's individual kinetic quantity. The three highlighted seam markers also cannot be equated to the source detector's thousands of would-flag events. Their selection appears to be different postprocessing, whose precise algorithm remains missing. [C79]

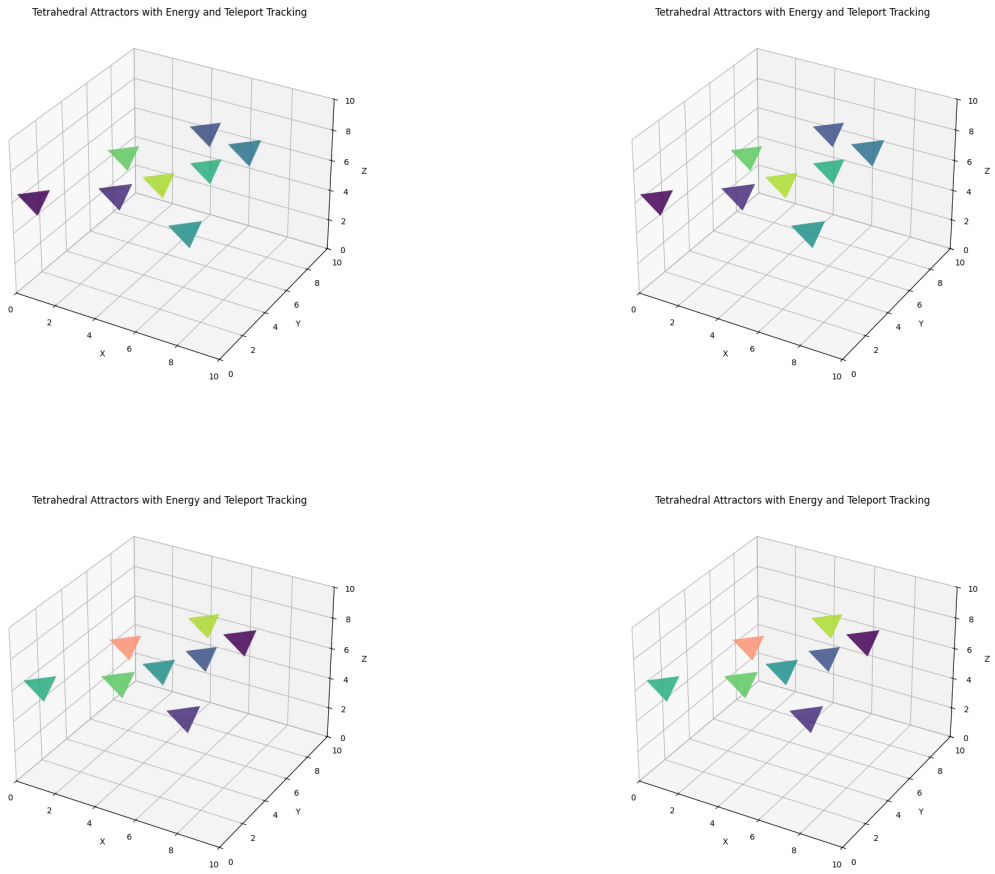


Figure 4.6: ORIGINAL movie content at left; DERIVED renderings of ORIGINAL saved B states at right. Top: movie frame 1 versus Bq2. Bottom: movie frame 6 versus Bq7. These are two of the eight fixed comparisons already performed in Study 03 using source geometry and current rendering defaults. No trajectories or fitted parameters were generated. Strong content agreement supports a candidate correspondence, while the original-run join remains formally UNLINKED. [7]

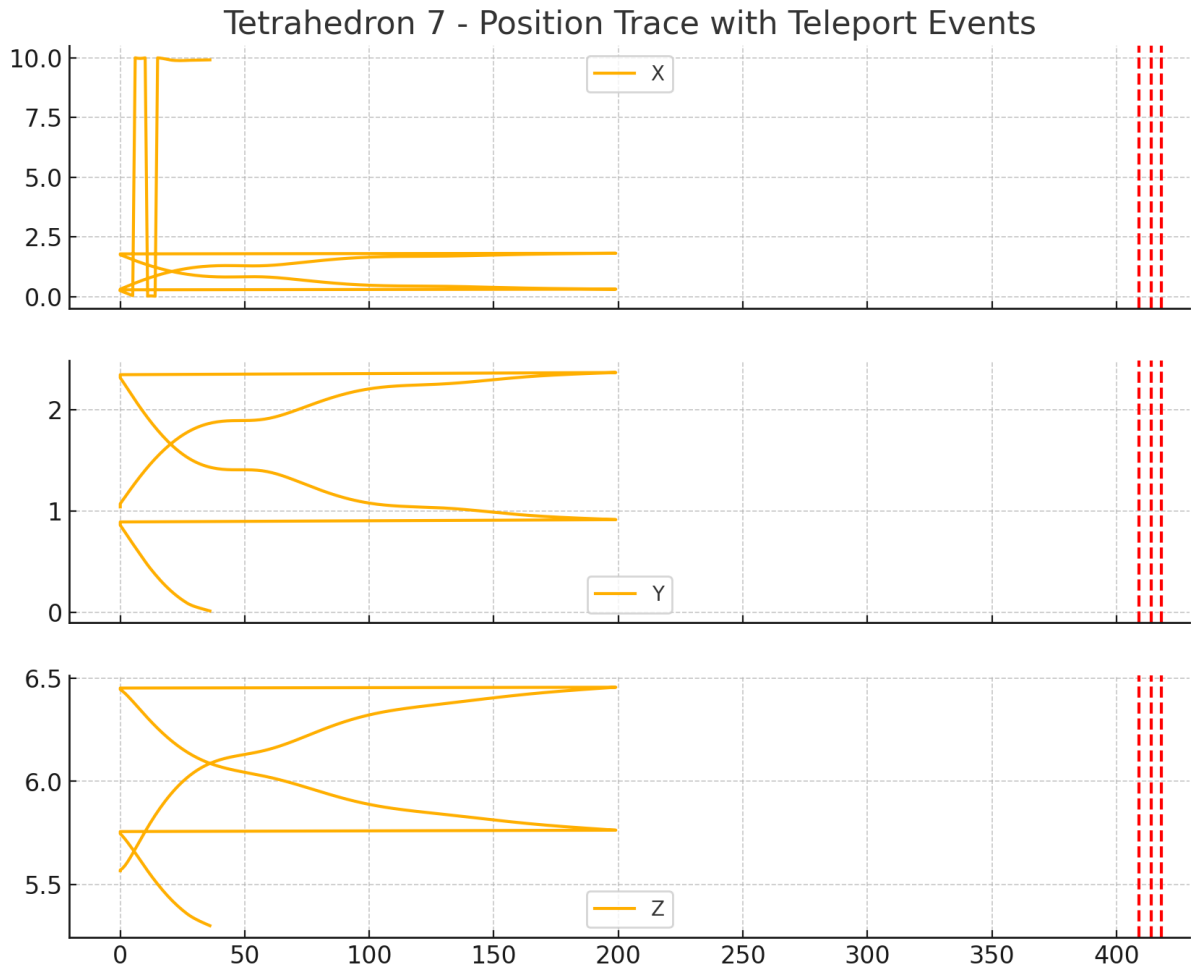


Figure 4.7: ORIGINAL historical chart / ATTRIBUTED to postprocessing of log B. Position traces use repeated Step labels 0–199, while markers 409,414,418 match callback-index seam events. The original PNG is unchanged. S03 contains no such PNG export and the chart generator is missing; exact plotting code and marker-selection algorithm are not certified. [7]

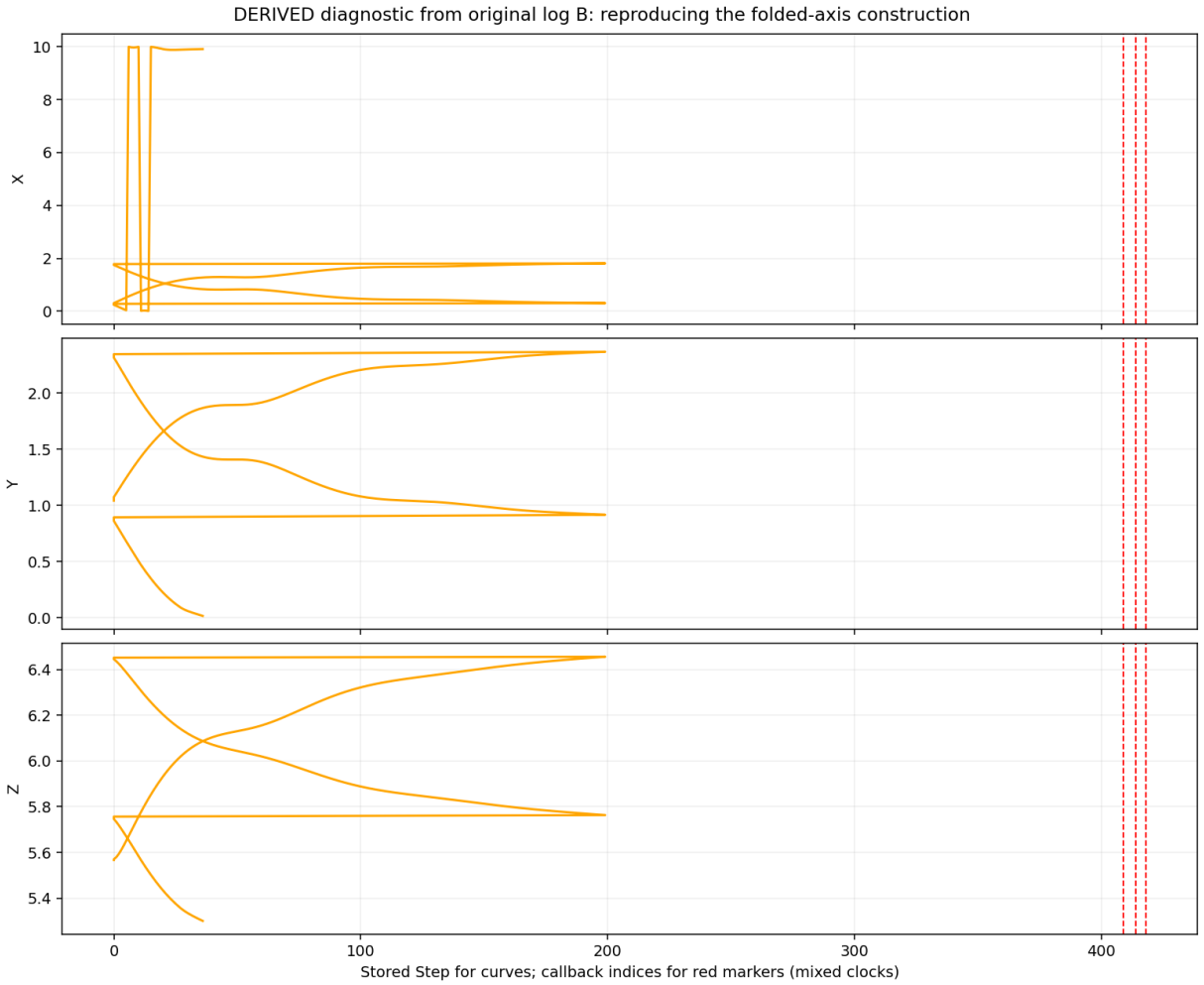


Figure 4.8: DERIVED from ORIGINAL log B, reused from Study 03. Repeated Step labels are used for the body 7 position traces while callback indices locate the three event markers. This recreates the misleading axis convention as a diagnostic; it does not repair the original chart or recover its exact generator. Lines joining label 199 back to 0 are drawing connections, not new trajectories. [7]

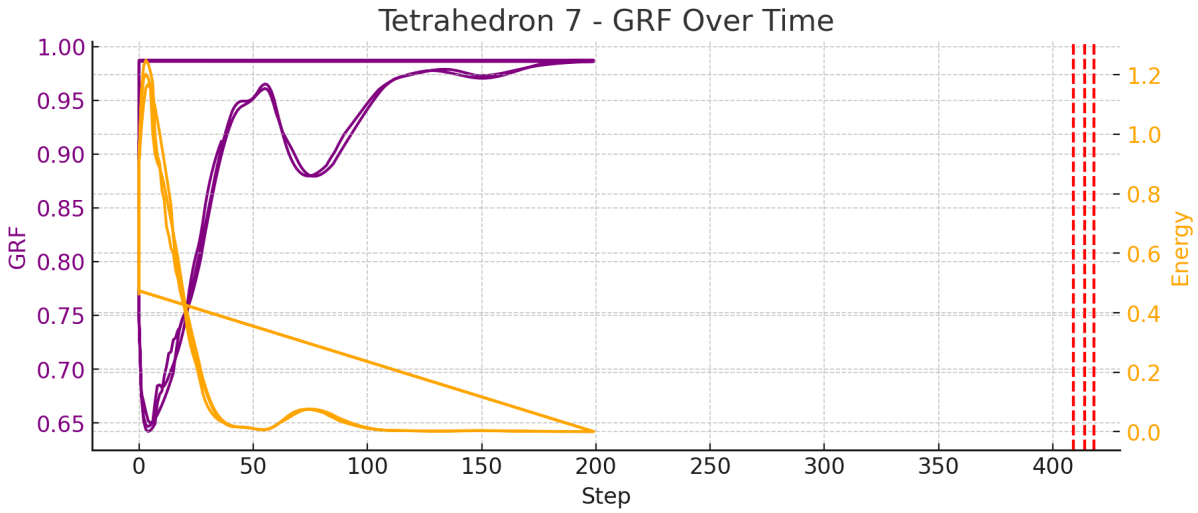


Figure 4.9: ORIGINAL historical chart / ATTRIBUTED to postprocessing of B. The body 7 title does not make the Energy column individual energy: the source repeats global K on every body's row. Repeated Step labels superpose separate passes, and event markers use callback indices. The exact chart generator remains missing. This preserved image is evidence of a reporting convention, not a conserved-energy measurement. [7]

Chapter 5

Shared mechanisms and genuine distinctions

In ordinary language. The programs share some formulas, but they are not one interchangeable experiment. Three pairwise-interacting points, eight centroid-coupled centers and eight centers with stochastic response signs have different states, interaction calculations and recording systems. Their evidence must be compared at the same level of certainty.

5.1 A precise mechanism comparison

Table 5.1 summarizes the source-supported distinctions. The complete side-by-side source locators are also supplied in `CROSS_STUDY_MECHANISM_MAP.md`. Similar helper names do not justify replacing one interaction law with another. [C80]

Table 5.1: Mechanism map. State dimensions count real position/velocity entries; the binary modes and external RNG/frame inputs are stated separately. Source support: S01 lines 5–74; S02 lines 8–119; S03 lines 10–167.

Property	S01	S02	S03
State	12 reals	48 reals	48 reals + 8 bits
Positions	3 in 2D	8 in 3D	8 in 3D
Initial velocity	Zero	Components in $[-0.1, 0.1)$	Same distribution as S02
Interaction	Two pairwise directions	Arithmetic centroid	Signed arithmetic-centroid response
Regularizer	Norm +0.001	Norm +0.01	Norm +0.01
Order	0,1,2 in place	0 through 7 in place	0 through 7 in place
Step multiplier	0.05	0.1	0.1
Coordinate period	2	10	10
H argument	Loop index $\times 0.05$	Frame label $\times 0.1$	Frame label $\times 0.1$
Speed feedback	Absent	Shared GRF	Same GRF, also toggle gate
Randomness after setup	Absent	Absent	Gated per-body draws
Stored history	Post-sweep position log	No numeric history export	CSV on every callback
Display	Static connected trails	Moving tetrahedra	Moving tetrahedra with mode palettes

S01's sum of regularized pairwise directions and S02's direction toward an arithmetic mean

are different nonlinear constructions. Averaging positions before normalizing is not the operation written in S01. The change of dimension, boundary period, regularizer and initial velocity distribution further prevents interpreting their pictures as a controlled comparison of one isolated mechanism. S01, lines 45–56; S02, lines 85–96. [C81]

There is a narrower exact relationship between S02 and S03: given identical incoming numerical arrays and an identical frame label, if every S03 post-gate mode is zero, its position/velocity update is the same mathematical update as S02's. Both share the same mixed-state centroid, H, GRF, norm regularizer, RPCO and wrap. This conditional identity follows by substituting $b_i^+ = 0$ into Equation (4.3). It does not establish equal historical initializers, random streams, callback schedules or trajectories. S02, lines 74–104; S03, lines 100–149; [7], SIMULATION_02_COMPARISON. [C82]

5.2 Sequential order is part of each model

All three programs overwrite positions before later bodies are processed. Consequently the mathematical state seen by a later interaction is a mixture of new and old rows. Equations (2.4) and (3.3) preserve that dependence. A simultaneous proposal/commit implementation would generally calculate different directions; it would be a modified model, not an innocent translation of the historical source. [C83]

The pictures show end-of-sweep states, so they hide this internal order. S01's accepted replay supplies all within-sweep arrays and demonstrates the dependency directly in events 1–3 and 505–507. S03's CSV exports only complete callbacks; its mixed-state values used in accepted residual checks are reconstructed from observed adjacent blocks. Those different observation capabilities must remain visible in the method, even when the algebraic notation is similar. [C84]

5.3 Persistent velocity and componentwise damping

For any scalar component u_a, Equation (2.3) gives

$$\operatorname{sgn} \mathcal{R}(u)_a = \operatorname{sgn} u_a, \qquad |\mathcal{R}(u)_a| \le |u_a|, \tag{5.1}$$

with strict magnitude reduction when $u_a \neq 0$. This follows because $1 + \log(1 + |u_a|) > 1$ for a nonzero component. Different component magnitudes receive different denominators, so the vector direction can change. The magnitude comparison is with the *candidate* velocity after adding the drive, not necessarily with the previous stored velocity. S01, lines 29–31,54–56; S02, lines 44–46,91–96; S03, lines 47–49,123–126. [C85]

Thus an inward response does not imply inward movement during the same update. S01's near-contact and S03's delayed switch response are two distinct observed examples. A zero instantaneous interaction also does not erase a carried nonzero velocity. No source implements a waiting state or a threshold that freezes every small velocity. These source facts are not a theorem against approximate waiting or approximate collective following over a particular finite window. [C86]

5.4 A prescribed gain is not a route memory

For nonnegative t, the shared H formula satisfies the elementary bound

$$|H(t)| \le 2e^{-0.5t} + e^{-0.3t} + 1.5e^{-0.7t} + e^{-0.4t} \longrightarrow 0 \quad (t \to \infty). \tag{5.2}$$

This is the triangle inequality applied to Equation (2.1), using $|\sin t|, |\cos t| \leq 1$. It does not say H is positive or that its magnitude decreases at every sample. The accepted S02 analysis evaluates $H(3\pi/2) \simeq -0.03600350414$ as a formula example; that time is not claimed as a nominal frame label. [6], TIME_AND_MOVEMENT. [C87]

The bound concerns increasing arguments. It does not make a restarting frame clock decay forever as callback count increases. S03's original records demonstrate why that distinction matters: at each return to label 0 the prescribed gain returns to 4.5. Neither H nor GRF reads a saved itinerary. GRF is state feedback through current mean speed; carried velocities are another form of state persistence. A stored route that explicitly guides future movement is absent from these three sources. [C88]

In ordinary language. There are several meanings of "memory" here. The current velocity retains effects of the past. A log records the past for an observer. H is a preset function of a supplied clock. These roles should be named separately before relating them to a broader SRG time theory.

5.5 Coordinate geometry and measurement

The coordinate rules identify opposite edges but do not add a material container. Minimal wrapped displacements, native increments, stored coordinate differences and screen distances are separate quantities. S01 draws connectors between wrapped samples; S02 and S03 draw current shapes without trails. S03 adds a diagnostic whose unsigned modulo differs from the displacement wrap used in the force. [C89]

The arithmetic centroid has its own coordinate-representative dependence. A seam crossing can change the interaction target even when the native movement is small. The measured S03 body 7 episode is therefore not adequately described by calling every visible effect a rendering artifact. Conversely, a large screen relocation alone does not certify a large native movement. Equations (3.3), (3.8) and (4.10) make the three operations distinguishable. [C90]

5.6 What the experiments establish together

The evidence supports three bounded conclusions. First, source order and carried velocity explain why an interaction direction and a movement direction can differ. Second, callback history and clock labels are necessary for interpreting a time-dependent gain: a 200-frame movie is not automatically a 200-update record. Third, the recording and rendering conventions can either reveal or hide the distinction between near-contact, projected overlap, seam crossing and a detector warning. [C91]

It does not follow that the stochastic mode is responsible for every visible difference between the two eight-body movies. Their unknown initializers, distinct nominal durations, blitting settings and encoding rates are uncontrolled differences. Nor do the finite records establish asymptotic convergence, a persistent orbit, a universal collective-following phenomenon or a physical theory of gravity. The kinetic bookkeeping identity describes the code's diagnostic; it is not a gravitational energy equation. [C92]

Chapter 6

Open questions and the next volumes

In ordinary language. The first volume gives us a reliable account of three constructions. It also makes the missing pieces more precise. Future work should begin with a source or record that can answer a stated question, rather than trying to reproduce a remembered picture. No further experiment is started or authorized by this chapter.

6.1 Recoverable identities and execution histories

The highest-priority provenance questions concern the run behind S01's historical PNG, the numerical history behind S02's movie, and a source/RNG/export chain for S03's movie and logs. An export receipt, saved initializer, original environment record or ordered trajectory could change the level of attribution. Without such evidence, image similarity remains corroboration rather than a certified historical join. [C93]

S03's observed repeated labels already establish that numerical updates and encoded frames are different counts. The missing historical GUI event trace leaves the cause of each pass unresolved. For S02, even the full callback count is missing. A future study could examine historical environment records or a separately approved bounded reproduction of callback behavior, but this volume performs neither. The current Matplotlib excerpts document a concrete compatible implementation, not a recovered historic software version. [C94]

6.2 Original time concepts and equation lineage

The source-defined time arguments are now explicit: a direct loop index in S01 and supplied animation labels in S02/S03. The broader conceptual origin of H, the intended interpretation of the names REFU/TGMO/RPCO/GRF, and their relationships to other SRG time equations require a separate lineage investigation. The implemented functions alone cannot settle their historical motivation or a physical theory of time. Study order and directory numbering are not substitutes for dated source evidence. [C95]

6.3 Path inheritance as a question about a chronology

The remembered chronology in which one component travels first and a later close pair follows an earlier route remains a research question. It requires identities, ordering, a specified finite window and a distinction between approximate following and exact coincidence. An accumulated static line picture supplies too little of that information. The seed-0 near-contact does not establish the remembered episode. [C96]

The earlier compatibility and initializer investigations are separate limited mathematical analyses of specified branches and conditions, as retained by the research charter. They are not recovered historical trajectories, a universal no-go theorem for SRG, or a reason to reject approximate collective behavior when exact coincidence fails. This volume neither extends those proofs nor tests another initializer. [8, 10] [C97]

6.4 Order, wrapping and attribution of causes

The source reconstruction identifies order dependence, arithmetic wrapped-coordinate averaging and display conventions as mechanisms that can alter interpretation. Separately approved future work could quantify their effects in a controlled comparison. Such a comparison would have to distinguish the historical rule from an explicitly modified diagnostic model, retain a known initializer and clock, and avoid presenting alternative rules as restoration of lost source. [C98]

For S03, a full audit of every velocity transition would increase verification coverage beyond the accepted sample, but is not performed here. Recovering the two body 7 chart generators would also resolve their exact event-selection and horizontal-axis conventions. The old charts are retained with a qualification, not silently corrected or replaced. [C99]

6.5 Later families and the gravity question

Later geometric, amplitude-coupled and neighbor-change simulation families remain outside Volume I. They should be reconstructed one source at a time using their own state, interactions, evidence and execution history. Fixed-grid models, glyph/echo narratives, March engines and production kernels must not be combined into one universal recurrence merely because terms recur. [8] [C100]

A future comparison between attractor response, geometry and gravity-related energy quantities may be scientifically useful. It begins as a question about exact existing equations and their domains, not as a result of the present kinetic diagnostic. Nothing here identifies the coordinate measuring framework as a physical container or force, adds a missing gravitational law, or makes a physical conservation claim from the movie. [C101]

6.6 The review boundary

Volume I is a substantial first reconstruction of three selected programs, not completion of the SRG project. Scientific and editorial review is complete. Subsequent experiments and GitHub publication remain separate decisions. This edition preserves the accepted science. Publication preparation does not itself authorize staging, commits, pushes, releases or another experiment. [C102]

Appendix A

Source identities and evidence locations

A.1 Immutable program identities

All paths in this appendix are relative to the repository's `SRG_REDISCOVERY` directory unless stated otherwise. The manuscript includes byte-identical small source copies in `MANUSCRIPT/sources`; the originals and accepted study packages remain unchanged. Line numbers in the text refer to these unchanged programs, not to generated listings.

S01: recursive_attractor.py, 74 lines, 2,074 bytes
`OLD_MATERIAL/Simulations_Trash/Simulations_part_2/recursive_attractor.py`
SHA-256: `847819ec381e200d9008395175342ec138e52ab3180a4f599b4db1cb91ae42d3`.

S02: 3Dtriangles.py, 119 lines, 3,970 bytes
`OLD_MATERIAL/Simulations_Trash/Simulations_part_2/3Dtriangles.py`
SHA-256: `4c53678eefd803e93a910b74e7a8096bfd5095e0d09d09c9df0e7cf514b312f1`.

S03: plot_1.py, 167 lines, 5,711 bytes
`OLD_MATERIAL/Simulations_Trash/Simulations_part_1/plot_1.py`
SHA-256: `e13b92df8ac6df78cfee75b95dacea3b106e96bd5be9bf798cdcf39ccc5bc75c`.

No additional historical source aliases were confirmed in the focused recovered index used by the accepted studies. Package copies are preservation copies, not independent historical versions. None of the three sources declares a seed, author/date/version statement or complete export environment. Current file timestamps were not used to infer invention order.

A.2 Historical images, movies and logs

S01 historical image
`OLD_MATERIAL/Simulations_Trash/Simulations_part_2/attractors.png`
800 by 800 PNG, 49,967 bytes. SHA-256: `72776de5e6228f4c4a0e2ef43d804f479bf8db80af6884e492e9ccbc3053d651`. Original / attributed; seed, ordered log and saving method missing.

S02 historical movie
`OLD_MATERIAL/Simulations_Trash/Simulations_part_1/tetrahedral_attractors_animation.mp4`
1,002,986 bytes. SHA-256: `4f90ce9fc384b134b68b5a64424ef3e05d364153d4e70efdd1596c0b8c937dd3`.
2,000 frames at 60 fps. Source association strong; exact run and numeric history missing.

S03 historical movie
`OLD_MATERIAL/Simulations_Trash/Simulations_part_1/tetrahedral_attractors_logic.mp4`

176,442 bytes. SHA-256: 0289d8e5a084e53f536183ad87d83b4a4c5ab972851bb61922a699406623203a. 200 frames at 20 fps. Source association strong; exact movie/log join formally UNLINKED.

S03 original log A
OLD_MATERIAL/Simulations_Trash/Simulations_part_1/PLOT_1_tetrahedral_states_20250712_015421.csv
1,451,175 bytes. SHA-256: 6e6fc49be7dbb5ecf77aa1fbeaf5106df12c04ef3c63c29ee84c693f2ed70e74. 8,736 rows / 1,092 complete callback blocks.

S03 original log B
OLD_MATERIAL/Simulations/CSV/tetrahedral_states_20250712_015647.csv
581,988 bytes. SHA-256: 051f8ad8be54794fcf54171a316d9931cc65b43349634799f446c53eb1b9afe8. 3,520 rows / 440 complete callback blocks.

Log A has byte-identical aliases at OLD_MATERIAL/Simulations/CSV/PLOT_1_tetrahedral_states_20250712_015421.csv and OLD_MATERIAL/Simulations_Trash/Simulations_part_3/PLOT_1_tetrahedral_states_20250712_015421.csv. They are one evidence stream at three preserved locations. The two filenames suggest 2025-07-12 times but do not certify a timezone, source revision or movie export identity.

The two body 7 historical charts lie in OLD_MATERIAL/Simulations_Trash/Simulations_part_1. The position chart has SHA-256 bf6dfb6ec1da4a9ceb56c9107b654671492576945a1b62426625b18ec99fe798; the GRF/energy chart has SHA-256 8f3c93d74ec1c03f97d70320b5c760953dca805b3bd3f1e967d54d6789a5b137. Their full names, copy hashes and unaltered images appear in the figure register. Both remain attributed postprocessing, with generator code missing.

A.3 Accepted packages and reproduced records

The authoritative package roots are:

- SRG_SIMULATION_CONSTRUCTION_01;
- SRG_SIMULATION_REPLAY_01;
- SRG_SIMULATION_CONSTRUCTION_02;
- SRG_SIMULATION_STUDY_03_PLOT1.

Each retains its own manifest, detailed scope and review-stage records. The new volume does not rewrite those receipts.

Inside the replay package, replay/trajectory.npz preserves the binary state arrays, SHA-256 b14901adc4612a3e557615fd0833f58ccb953d3f84f8823f1121a5bef5b0c5e1. replay/states.csv has SHA-256 4cdc156fd4cc777220528805a85dee65298b8c44b723e3766a65809ffb6b6ff9. replay/within_sweep.jsonl has SHA-256 af43911ab0dc3fa7bda5ae96e418ccac1e93da30839124abeb6a3514cb4b80ff. These are reproduced records from the known seed, not original historical trajectory data.

The volume's external evidence index and figure/media register provide relative links into these packages. They avoid duplicating every movie, frame, CSV and prior ZIP. A detached review ZIP contains the manuscript, its selected figure assets, source copies and supporting review records; the complete numerical/movie evidence is accessed through the stated repository-relative locations.

A.4 Excluded and qualified evidence

The curvature-labelled CSV previously indexed by R0 belongs to a different, incompletely recovered rule. It is not evidence for S02 or S03's trajectories here. The nearby CRE/DRE and

harmonic-tunnel images inspected in Study 03 contain quantities or narratives absent from the selected program; they are not used as behavior evidence. No neighboring movie or memory-based description substitutes for an unrecovered run.

The chart-axis and detector findings qualify how historical outputs should be read; they do not alter those outputs. The wording “delayed direction reversal” in this manuscript refers to the recorded velocity relative to a source-defined centroid direction, without claiming physical calibration. The signed response mode is likewise not an invariant label for attraction whenever H changes sign.

Appendix B

Inspection references and review reproducibility

B.1 What can be inspected without another experiment

The accepted packages contain the evidence required to audit this account without advancing a simulated state. S01's source and original image establish its construction and historical display. Its replay package supplies initial state, run specification, complete state records, selected event windows, fidelity receipt and saved-record checks. The replay harness is preserved for provenance, but this volume does not instruct its execution.

S02's package contains the unaltered movie, selected full-resolution frames, inspection sheets, metadata, frame-count receipt and Matplotlib source excerpts. S03 adds the two complete original logs, callback tables, switch/wrap event tables, pair minima, sampled equation residuals, independent-check receipt, episode records and content-correspondence results. Its inspection scripts are references for how the accepted results were obtained; they were not rerun during manuscript preparation.

Table B.1: Verification coverage retained from the accepted studies. The present volume performs editorial and preservation checks only.

Evidence	Accepted check	Important limit
S01 reproduction	500 position snapshots and final velocities bitwise equal to unchanged copied-source execution	No lost historical-run recovery; intermediate reference velocities not compared at every sweep
S02 movie	All 2,000 frames decoded; selected sheets/stills visually inspected	No full numeric trajectory; no certified native displacement from pixels
S03 logs	All energy, position and GRF relations; 31 sampled full velocity transitions; independent 20 R0 samples	No first pre-log state; missing RNG draws; not every velocity equation audited
S03 movie/log candidate	All 18 recorded zero-label candidates versus movie frame 0; eight fixed $f \leftrightarrow q = f + 1$ pairs	Descriptive pixel evidence; exact original-run join remains UNLINKED

B.2 Manuscript build

The professional source is LaTeX: `MANUSCRIPT/volume_i.tex`, its section files, references and figure includes. The candidate PDF is built from that source, not from an unrelated text export. A pinned Tectonic 0.17.0 binary was obtained from its official release for this draft, with the download checksum matched to the release asset digest. Its provenance and executable hash are recorded in `records/typesetter_provenance.json`. The toolchain and cache remain inside the draft's build directory and are excluded from the review archive.

Build and inspection commands use Windows Command Prompt with the requested conda environment. From the repository root:

```
call C:\Users\Notandi\miniconda3\condabin\conda.bat activate torment
```

Then change to `SRG_REDISCOVERY/SIMULATION_RECONSTRUCTION_RESEARCH/VOLUME_I_01_R1`. In a fresh extracted Windows copy, first obtain the pinned compiler, then run the manuscript-only builder:

```
python -B tools\acquire_typesetter.py
python -B tools\build_pdf.py
```

Existing workspaces with `build/toolchain/tectonic.exe` already present can skip acquisition. First-time dependency acquisition needs network access. The compiler and build cache are excluded from the review ZIP; the first build may also fetch uncached TeX dependencies. An alternative compatible XeLaTeX setup is a separate build route; cross-environment byte identity is not promised.

The builder executes no historical source, simulation harness or scientific checker. It sets a task-local Tectonic cache and temporary paths. The candidate's actual compiler receipt, PDF hash, page count and rendering review are retained in the review records.

B.3 Figures and presentation

All scientific figure content comes from accepted packages. Two replay sheets are cropped to their already-recorded four-panel rows and arranged in a readable two-by-two layout; crop rectangles and input/output hashes are explicit. Original and derived movie stills are placed side by side only for their already-tested correspondence, never joined into a combined trajectory. Other chosen images are byte-identical copies scaled on the page.

Each caption states evidence class, origin, axes or frame indices, and a limit on inference. The figure register lists every asset's SHA-256, source path and transformation. No scientific data were replotted or recalculated for a more attractive outcome.

B.4 Preservation and acceptance

The initial preservation snapshot covers 397 files: the four accepted packages, including their archives; the previously protected originals, aliases and supplied instructions; the canonical charter and the Volume I work order. The canonical ledger is separately captured before its explicitly authorized minimal update. This is exact scoped coverage, not a claim to hash every unrelated untracked file in the repository.

Git HEAD and index are recorded, as are all 220 initial status entries. The reconstruction workspace was already untracked at that level, so a new nested volume need not create an additional top-level porcelain entry. Preservation requires the prior entries to remain unchanged,

protected file hashes to match and the ledger diff to contain only the authorized acceptance and draft-navigation update.

The manuscript's claim matrix, figure register, mechanism map, open-question document, editorial checks and archive manifest are review deliverables. Build caches and the candidate's temporary page rasters are excluded from the compact review ZIP; the visual-review receipt describes the inspected pages. No staging, commit, push, upload or publication is part of this assignment.

Bibliography

[1] Historical program, *recursive_attractor.py*, 74 lines. S01 source identity and exact location in Appendix A. No declared seed or source date.

[2] Historical program, *3Dtriangles.py*, 119 lines. S02 source identity and exact location in Appendix A. No declared seed or source date.

[3] Historical program, *plot_1.py*, 167 lines. S03 source identity and exact location in Appendix A. No declared seed or source date.

[4] GPT scientific lead; Codex investigation. *SRG Simulation Construction 01*. 22 October 2026. Accepted baseline: START_HERE, CONSTRUCTION_WALKTHROUGH, MANIFEST, preserved source and historical attractors.png. `SRG_REDISCOVERY/SRG_SIMULATION_CONSTRUCTION_01`.

[5] GPT scientific lead; Codex investigation. *SRG Simulation Replay 01*. 22 October 2026. Accepted seed-0 reproduction: MOTION_WALKTHROUGH, REPLAY_VALIDATION, RUN_SPEC, initial state, execution/equivalence receipts, numerical records and figures. `SRG_REDISCOVERY/SRG_SIMULATION_REPLAY_01`.

[6] GPT scientific lead; Codex investigation. *SRG Simulation Construction 02*. 22 October 2026. Accepted construction, TIME_AND_MOVEMENT, VISUAL_EVIDENCE, comparison, manifest and movie inspection records. `SRG_REDISCOVERY/SRG_SIMULATION_CONSTRUCTION_02`.

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